Higher Scores, Not A Better Model: A Large Multiple-Choice Lift That Is Mostly Inducible, Off-Axis, and Does Not Transfer

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Abstract

Contrastive-correctness parameter-efficient fine-tuning (the LoFiT/ITI family) is widely used to sharpen a model's multiple-choice judgment cheaply. On a frontier 31B instruction model (Gemma 4 31B IT) it buys a large cold multiple-choice lift: ARC-Challenge +35.0, HellaSwag +17.0, Winogrande +19.0 (scoring-face, no chain-of-thought), plus TruthfulQA-mc1 +37.25, which we cite separately throughout as an answer-prior selection lift and exclude from the mechanism. We ask what that lift *is*, and answer with a quantified causal attribution rather than a new method.

Most of the lift is *generic to fine-tuning*. A capacity- and budget-matched non-contrastive control recovers ~62% of the ARC lift (fraction-CI [45%, 85%]; flat across rank r8–256, and a best-effort r512 control does not raise it, so ~62% is a tight lower bound). That places the lift on a matched-control spectrum of *three* rungs — random ≈0% → generic SFT ~62% → contrastive 100% — with an on-axis reference (W_{know}) at 6.4%, *statistically indistinguishable from zero* (exact McNemar $p = 0.136$), counted with the floor rather than as a fourth step. To our knowledge no prior work runs a matched control and reports this *fraction*; that attribution is the load-bearing novelty.

The recovery is *causally off-axis of degree*, and the identifying evidence is the negative arms: a push *parallel* to the correctness direction recovers 0% of the lift, and a norm-matched *random* perpendicular push recovers ~0% across three seeds — so the learned perpendicular direction is privileged, not any perturbation of the right size. (A perpendicular push recovers ~100%, but the offset is already almost entirely perpendicular — $\cos(\theta_{\text{mc}}, v_{\text{inf}}) \approx -0.003$ — so that arm is near-tautological and we do not lean on it.) The lift is also *undeployable as a generator*: it does not transfer (gsm8k −25 pp, a mechanistic multi-step-reasoning loss; MMLU −4.2 pp spillover; out-of-task, GPQA cold-MC ~0 while the base reasons GPQA out at CoT 0.70, single-run and directional). A per-item arbiter bounds it: the base already solves 124/124 of the fixed ARC items under chain-of-thought, against 41/47 on the items the adapter fails to fix ($p = 1.4 \times 10^{-4}$) — the arbiter discriminates rather than restating a ceiling.

A *read-only* linear selector recovers the lift (ARC selector 0.943 across the full n=400 evaluation set, not only base-wrong items) without touching generation, Pareto-dominating the adapter — the read leg of a read≠write dissociation (the selector capability itself is prior work, NoVo/CORAL, used here as evidence, not claimed as new). We likewise build on, rather than lead on, the established fact that cold-MC scoring under a chat template is a lossy readout of what the model knows (Confidence-Manifold, Inside-Out, KAPPA). Both the effect and its off-axis geometry replicate cross-family (Qwen 14B: same-sign lift on all four tasks, three clearly and Winogrande marginal at n=400; whitened $\cos(\theta_{\text{mc}}, v_{\text{inf}})$ −0.002, 97.5% perpendicular energy). Practically, this is a caution for the contrastive-PEFT-for-multiple-choice program: a cold-MC lift of this size can be an off-axis, non-transferring readout edit — matched by a cheaper read-only selector, and dominated at its own single-forward cost by the most trivial same-cost baseline, simply asking the base to emit the answer letter (0.970 on base-wrong items vs the adapter's 0.763, tying a reasoning baseline that costs ~1000× more). A benchmark gain of this kind should not be read as a capability gain. This is a mechanistic study of one observable, not an architecture.

1 Introduction

Contrastive-correctness parameter-efficient fine-tuning — the family that learns per-head additive offsets from contrastive pairs of correct and incorrect answers, of which LoFiT [42] and inference-time intervention (ITI) [22] are representative — is an attractive way to raise a model's multiple-choice score at near-zero parameter cost. Applied to a multiple-choice discrimination objective on a frontier 31B instruction model (Gemma 4 31B IT), it does exactly that: cold multiple-choice scoring on ARC-Challenge rises by +35.0 accuracy points and on TruthfulQA mc1 by +37.25, with gains on HellaSwag (+17.0) and Winogrande (+19.0).

A benchmark movement of this size invites the conclusion that the model has become better — that the adapter has *sharpened the model's judgment*. It has not. The lift does not transfer to generation (gsm8k −25 pp). On the items it fixes, the base model already reaches the right answer by reasoning (124/124 under chain-of-thought). A read-only linear selector on the same activations recovers the lift without touching the weights or the decoder. And the most trivial baseline imaginable beats it at its own cost: asked to emit the answer letter in a single forward, no chain of thought, the base recovers 0.970 of the base-wrong items against the adapter's 0.763 — tying a reasoning baseline that costs a thousand times more. The score moved and the model did not.

This paper asks what that lift *is*, and answers with a quantified causal attribution rather than a new method: most of it is what ordinary fine-tuning already buys, it lives in a learned direction that is off-axis to the model's own correctness geometry, and it re-ranks a lossy readout rather than adding capability.

Most of the lift is inducible by ordinary fine-tuning. If the lift were a specific property of *contrastive-correctness* training, an ordinary, non-contrastive supervised fine-tune of matched capacity and budget should not reproduce it. It largely does. A matched Align-LoRA — plain SFT on the same data budget, run through the same causal recovery machinery — recovers about **62%** of the ARC lift, and that fraction is flat across a rank sweep (r8/32/64/256: 59/63/60/65; the slope's confidence interval includes zero, so the residual is not a capacity gap the control can close).

Read correctly, this matched control *is* an attribution method, and the 62% is the result, not a deflation. It spans a series — a norm-matched random direction ≈0% → generic SFT ∼62% (fraction-CI [45%, 85%]) → the contrastive adapter 100% — that places most of the lift on fine-tuning *in general*, not on the contrastive objective, leaving a contrastive-specific remainder of at most ∼38%. The strongest on-axis reference we can build (a Fisher-LDA direction W_{know}) sits at 6.4%, but its interval includes zero (exact McNemar $p = 0.136$), so we count it with the floor rather than as a distinct rung: the series has three steps, not four. That remainder is an upper bound: a best-effort fairness control has since run, and raising capacity to r512 recovers *less* (0.22) or breaks below base, so it does not raise the generic fraction, and the ∼62% stands as a tight lower bound.

The clean test is ARC; TruthfulQA is excluded because its matched corpus leaks into the evaluation (held-out, the capacity-matched control collapses from 98% to 30% recovery — the signature of memorizing train-on-eval — while the contrastive adapter still generalizes held-out at 0.83).

Neighboring work establishes the recognition–production gap qualitatively (Section 2); to our knowledge no prior work runs a *matched* control and reports the recovered *fraction*, which is the load-bearing novelty.

The recovery direction is off-axis to correctness — off-axis of degree, and established causally. Where the recovered lift lives relative to the model's own correctness geometry is not a matter of inference from weights but of a direct causal test. Pushing an offset *perpendicular* to the functional correctness direction recovers ∼**100%** of the lift; pushing it *parallel* recovers **0%**; and a norm-matched random direction inside the same perpendicular complement recovers ∼**0%** across three seeds. Of these three arms, the perpendicular one is the least informative: because the offset's cosine with the correctness direction is already ≈ − 0.003, the perpendicular component *is* essentially the whole offset, so its ~100% is close to a restatement (Section 5.5). The weight falls on the other two — an on-axis push at the same scale does nothing, and an equal-norm random direction in the same subspace does nothing either.

Two facts follow at once. The lift is genuinely off the correctness axis — an on-axis push does nothing — yet the *learned* perpendicular direction is privileged, since an equal-norm random perpendicular direction does not reproduce it. This closes the standard objection that an off-axis direction is unidentifiable because *any* norm-matched perturbation would work: it is closed on the harness side (the same machinery that yields 100% for the perpendicular push yields only 6.4% for the on-axis reference, so the on-axis failure is real, not an injection artifact) and on the write side (the learned perpendicular push vs the shuffle control).

Comparing the net activation effect of each method, the geometry is a *method-dependent spectrum*, not a single fact: both the contrastive adapter and generic SFT are mostly off-axis (cosines 0.088 and 0.199 with the correctness direction), but generic SFT is 2.3× more on-axis.

We apply a caution to ourselves here — a zero-cost weight-space preview reported "off-axis" (ratio ~1.05) and the clean activation experiment refuted it (ratio 3.93); we report only the clean experiment.

Cold-MC scoring under a chat template under-expresses what the model knows — established setup we build on. That cold multiple-choice scoring under a chat template is a lossy readout of a model's knowledge is, by now, documented across model families with causal validation: internal probes recover correctness at 0.80–0.97 AUC where output-based methods reach only 0.44–0.64 [11], the internal-vs-external knowledge gap averages 40% [15], and the residual stream carries distinct knowledge and prediction subspaces [29]. We take this as setup, not as our contribution, and confirm it holds in our channel with a per-item arbiter.

When we take the 124 ARC items that *only* the adapter fixes in cold scoring and ask the base model to answer them by generating a chain of thought, the base solves 124/124 (1.000) — indistinguishable from its CoT accuracy on items it already scores correctly cold (200/202 = 0.990). What makes that informative rather than circular is the other side of the same run: on the items the adapter *fails* to fix, the base solves only 41/47 (0.872, $p = 1.4 \times 10^{-4}$). The arbiter is therefore tracking which items the adapter moves, not restating that ARC is easy under chain-of-thought. The lift is not new capability; it is a re-ranking of cold scores toward answers the base can reason its way to but that no-reasoning scoring misranks.

This is why the lift does not transfer to generation (gsm8k loss in every condition), and the no-transfer is not an artifact of ARC being easy: on GPQA, where the base must genuinely reason and does (CoT accuracy 0.7071, well above chance), the cold-MC lift is itself ~0 (base 0.4747 → adapter 0.4697, CI crossing zero), so a scoring shortcut has no generative headroom to add even out-of-task.

(How the readout misranks on ARC — confident burial of the gold under fluent surface forms — and how far the present-but-swamped signal survives, we characterize in Section 5.1; here it is the established backdrop against which the *attribution* is the result.)

A read-only selector recovers the lift without the pathologies — the read leg of read≠write, on established setup. Because the correctness signal is present in the activations and merely under-expressed in the cold output scores, it can be read off directly, with no weight edit at all. That a read-only probe or readout can *select* MC answers and recover an accuracy lift is itself established — NoVo selects answers by voting over truth-correlated attention-head norms (+19 points on TruthfulQA-MC1) [19], and CORAL re-scores per-option probabilities with a frozen correctness probe (+14% on ARC-Challenge/HellaSwag) [27] — so we treat the read-only selector as setup, not as a new method.

Our use of it is as the *read leg* of a read≠write dissociation: a read-only linear selector on the same per-head activations the adapter writes to recovers the ARC lift (selector accuracy **0.943**, three seeds; fixes 181 base-wrong, breaks 5 base-right, exact McNemar $p = 3.7 \times 10^{-47}$) *without touching generation*, so it carries none of the adapter's deployment costs (no generation loss, no retrieval spillover) and Pareto-dominates the *adapter* on the (cost, MC-accuracy) plane. Unlike NoVo/CORAL — which present such a selector as a calibration/accuracy *fix* — we use it as causal evidence that the lift the adapter writes off-axis is read-recoverable on-axis without the write's damage.

We are precise: 0.943 exceeding the adapter's 0.853 is *not* "the probe beats the adapter as capability" (different information sets — 12288 activation dimensions vs collapsed output logits); it is a read-side statement that the cold-MC channel discards recoverable signal. Its feature set is held out by item but not by feature (the 48 cells are selected on the training split), so cross-benchmark generalization is the natural strengthening.

Two failure modes, reported separately. The two headline tasks do not fail the readout the same way, and we do not blend them. ARC, HellaSwag, and Winogrande carry the question-conditional readout story (the ARC lift retains 78% under a PMI / decision-conditional control; the choices-only lift is only +0.0375, CI crossing zero).

TruthfulQA is a second, *answer-prior / surface-form Goodhart* mode: the choices-only lift is +0.2225 and the adapter selects the correct option 0.5725 of the time with no question at all. The TruthfulQA gain does generalize held-out (Δ_{UNSEEN} mc1 +37.8 $\approx$ Δ_{SEEN} +36.9, so it is not memorization — and a generic Align control collapses held-out to +7.9, confirming the contrastive lift is genuine), but we read it strictly as *answer-prior selection on the mc1 metric*, never as a truthfulness gain and never as evidence for the off-axis mechanism, which is anchored on ARC.

This is a mechanistic study of an observable effect, not an architecture. We make no claim about routing as a contribution and no claim about a redesigned model. The object of study is one measurable phenomenon: what a frontier model's cold-MC scoring misses, and what contrastive-correctness PEFT actually does when it recovers part of it. Where we describe a routed system at all, it is strictly a mitigation vehicle that sends generation-face traffic to the unmodified base; it is not advanced as a result.

Scope, discipline, and what is not yet validated. We characterize this on Gemma 4 31B IT and corroborate the discrimination signature cross-family and cross-scale where artifacts exist. We hold ourselves to a verify-before-claim discipline: every number traces to its source run or is marked, and we freeze no prose on un-executed evaluations.

Concretely, the headline scoring/gen magnitudes are pre-registered with kill-rules. The anti-floor 5-shot rule has executed and the lift survived. The gen-face canonical gsm8k sweep (R6) has now run: −25 pp, flat across the generation budget, so the loss is mechanistic, not a clipped chain of thought. The contrastive-specific residual ~38% is a firm upper bound: the Align-LoRA best-effort fairness control has now run and does not raise the ~62% generic fraction (r512 recovers 0.22 / breaks below base), so the ~62% is a tight lower bound. The earlier V7-mc end-of-turn figure (60→4%) is **retracted**: it and the analogous D.1 numbers came from a tokenizer bug that scored termination on the wrong turn-stop token, and the one clean re-measure shows no collapse (Section 5.7). The D.1 open-gen degradation is real but modest (social 96→76%, creative 68→12%), adapter-dependent, and off the V7-mc headline. A temperature/null-space account of the offset is reported refuted (Section 5.8), now confirmed by causal patching, not merely direct-logit attribution. We mark each in place rather than leaning on it.

Contributions. The object of this paper is a dissociation — a large multiple-choice score gain that is not a capability gain — and the contribution is the causal attribution that explains it, together with the deployment pathologies the recovery introduces.

- **Inducibility-as-attribution.** A capacity- and budget-matched non-contrastive Align-LoRA recovers ~62% of the ARC lift — fraction-CI **[45%, 85%]** — (flat across rank; a best-effort r512 fairness control *ran* and does not raise it → a *tight* lower bound, contrastive-specific remainder ≤ ~38%). Read as an *attribution spectrum* — random ≈0% → generic SFT ~62% → contrastive 100%, with the on-axis reference (W_{know}) at 6.4% indistinguishable from zero ($p = 0.136$) and therefore counted with the floor — most of the lift is what ordinary fine-tuning buys back. To our knowledge no prior work runs a matched non-contrastive control to *attribute* an MC lift this way and report the fraction. The interval is wide, and we read the claim at its resolution: most, not a specific percentage.

- **A causal off-axis-of-degree geometry, non-identifiability closed both sides.** A parallel push recovers 0% of the lift and a norm-matched random perpendicular push ~0% (three seeds) — so the off-axis direction is learned and privileged, not any perturbation of the right size. The perpendicular push recovers ~100%, but since the offset is already nearly all-perpendicular that arm is close to tautological (Section 5.5); the parallel and shuffle arms carry the identification. The net-effect comparison is a method-dependent spectrum (contrastive cosine 0.088 vs generic SFT 0.199), and the lift has no layer locus (it scales with offset magnitude, correlation 0.985); a training-time on-axis penalty does not rescue transfer (see Section 8), adding *necessity* to this sufficiency.

- **No transfer, on an established lossy-readout setup.** Cold-MC scoring under a chat template under-expresses knowledge the model holds — established [11, 15, 29] — and we confirm it (the fixed items are CoT-accessible, 124/124 on ARC, against 41/47 on the items the adapter does *not* fix, $p = 1.4 \times 10^{-4}$) and show the no-transfer holds *out-of-task* (GPQA cold-MC ~0 at base CoT 0.70, single-run) and on generation (gsm8k −25 pp). That generation cost is coupled to *how* the offset is injected, not to the lift (a generic SFT recovers the lift but preserves gsm8k 0.96→0.95); a previously-claimed end-of-turn collapse is **retracted** as an eos-token artifact.

- **The read leg of read≠write (built on established setup, not a new method).** A read-only linear selector on the same per-head activations recovers the ARC lift (accuracy 0.943; McNemar $p = 3.7 \times 10^{-47}$) without touching generation — Pareto-dominating the *adapter*, though a reasoning test-time-compute baseline dominates both (Section 5.4). That a read-only probe-selector recovers an MC lift is established [19, 27]; our contribution is its *use* as the read leg of a read≠write dissociation auditing an off-axis write neither studies.

- **Two clean negatives, briefly.** The offset is *not* a temperature/null-space mechanism (refuted causally) and *not* a token-frequency edit (frequency control closed); and the lift is task-heterogeneous — ARC carries the question-conditional mechanism while *TruthfulQA is a second, answer-prior failure mode* (choices-only +0.2225), generalizing held-out but reported separately, never as the off-axis mechanism.

Scope at a glance. So the reader carries the boundaries once rather than re-encountering them piecewise: the causal spine (arbiter, inducibility control, recovery spectrum) is single-model (Gemma 4 31B IT) and single-adapter (one training run, no re-train replicates), with the *effect* and its off-axis *geometry* replicated on Qwen 14B; the mechanism claims are anchored on ARC (TruthfulQA is a separate answer-prior mode; HellaSwag/Winogrande corroborate); every claim is protocol-bound to cold-MC scoring under a chat template (without the template there is no lift to study, Section 4.1); n=400 denotes the first 400 items of each lm-eval split, with a full-split run confirming every headline delta (Section 4.1); the headline anchors are low-CoT-headroom benchmarks by design (Section 7); and the routed system is a mitigation vehicle, not a contribution.

Notation. Full definitions in Section 3.6. **V7-mc** — the contrastive-correctness adapter under study (48 attention heads, additive per-head offsets). $\boldsymbol{\theta}_{\mathbf{mc}}$ — its learned offset (aggregate write). $\boldsymbol{v}_{\mathbf{inf}}$ — the per-head functional correctness direction (mass-mean estimate); $\boldsymbol{W}_{\mathbf{know}}$ — the whitened Fisher-LDA on-axis reference. $\mathbf{off}_{\parallel}$ / $\mathbf{off}_{\perp}$ — the offset's components parallel / perpendicular to v_{inf}. **Cold-MC scoring** (= *scoring-face*) — log-likelihood ranking of options under the chat template, no generation; **generation-face** — evaluation of produced text. **Recovery fraction** — (arm − base)/(adapter − base) on the task's headline metric (Section 3.6).

2 Related work

2.1 The recognition–production gap

A robust body of work establishes that what a model can *recognize* (rank correctly among options, or encode in its hidden states) outruns what it can *produce* (generate correctly), and that this gap is real rather than an evaluation artifact. The generative-AI paradox documents systems that can judge outputs they cannot themselves generate [38]; probing work shows models often encode the correct answer in their representations while generating an incorrect one [28]; the original inference-time-intervention work reports a large gap between a truthfulness probe's accuracy and the model's generation accuracy [22]; and recent evaluation work argues that answer-matching on free generation is a stricter, more faithful test than multiple-choice scoring [9]. Two recent results establish the specific, causal-and-geometric form of the gap that we take as setup: the correctness signal is recoverable by internal probes at 0.80–0.97 AUC while output-based readouts reach only 0.44–0.64, validated causally by activation steering (a learned direction moves error rates 10.9 pp where random directions do not) [11], and a formal internal-vs-external knowledge measure finds an average 40% hidden-knowledge gap across open-weight models [15]. We take this lossy-readout fact as *established setup and claim no priority on it* — it is the backdrop, not our contribution. Our per-item arbiter (the base solves under chain-of-thought 124/124 of the items the adapter fixes in cold scoring) confirms the gap in our channel, but our contribution is the *converse* direction and a *quantified causal attribution*: we show that a cheap fine-tune recovers most of the cold-scoring deficit along a learned, off-axis direction, and that the recovery does not transfer back to generation — a property of what the adapter does, not only of the gap's existence. Closer to our read-only selector specifically, the gap has already been *operationalized* as an answer-selection method: an activation-derived readout selects the multiple-choice answer and recovers an accuracy lift with no weight edit — by attention-head-norm voting [19] or a frozen correctness probe re-scoring per-option probabilities [27]. We claim no priority on that capability either; we use it only as the read leg of a read≠write dissociation (Section 5.4), where the novelty is the contrast with an off-axis *write* these works do not study. A complementary line *does* study a causal write, but in the generation face rather than our cold-scoring channel: instruction-tuned models often fix their answer before reasoning, and a linear probe on the pre-CoT last-token activation decodes that answer at ~0.9 AUC while steering along the probe direction flips the produced answer in over 50% of cases, exceeding orthogonal baselines [12]. This sharpens rather than blurs our claim: in that decision/generation face an answer direction is genuinely writable, whereas in our cold-MC scoring channel the correctness direction is decodable (0.955, Section 5.4) yet the on-axis write fails (W_{know} 6.4%, Section 5.5) and the off-axis write that does lift never transfers back to generation (Section 5.1) — so our read≠write dissociation is *channel-specific*, not a claim that answer directions are never writable. Its two steering failure modes when a wrong answer is induced — non-entailment and confabulation, i.e. reasoning from a false belief — also parallel the reasoning-chain breakage our contrastive injection produces on gsm8k (Section 5.7).

2.2 Multiple-choice scoring of instruction-tuned models is an unreliable readout

That cold multiple-choice scoring is a poor proxy for an instruction-tuned model's behavior is established, and we claim no priority on it. First-token / log-likelihood option scores are severely misaligned with the model's actual text answer — mismatch rates exceeding 60%, worst on models heavily fine-tuned on conversational or safety data [36] — and the text answer is the more robust quantity when the two disagree [35]. Cold MC scoring is further sensitive to option ordering and symbol identity [30, 44], is poorly calibrated without correction [43], and instruction tuning polarizes the answer distribution [13]. Two recent results are especially close to our framing: instruction-tuned models can struggle to emit the correct option symbol even when they know the answer [8], and the chat template itself inflates a model's confidence in "its own" answer in a way removable by a parameter-free reframing [31]. We build directly on this literature — the protocol (cold MC under a chat template) is a lossy readout, which we treat as *established setup rather than as our thesis* — and our contribution is past the diagnosis: we quantify how much of the deficit a generic fine-tune recovers (~62% on ARC, a matched-control attribution), locate that recovery geometrically (off-axis of degree, established causally), and show it does not transfer. Critically, and unlike a "the lift is a formatting artifact" reading, the recovered lift is *real* signal: it survives a same-scaffold PMI decomposition (ARC retains 78%), so the readout is genuinely leaving knowledge on the table, not merely mis-normalizing it.

2.3 The mirror in compression: the benchmark illusion

The closest single neighbor in the same channel (log-likelihood scoring vs greedy generation) comes from model compression. Pruned models can pass multiple-choice evaluation while failing to answer the same question in open generation; the correct answer is usually not erased but *demoted* in rank, reappearing under beam search, sampling, or one in-context example [37]. That is our phenomenon's mirror: there, MC scoring *overstates* usable capability (the answer is demoted but MC-recoverable); here, cold MC scoring *understates* it (the answer is CoT-accessible but cold-misranked) and a fine-tune *promotes* it in the MC channel without helping generation. The shared lesson — multiple-choice scores are not generation capability — is established; the constructive, opposite-signed instance, and its off-axis mechanism, are ours.

2.4 The knowledge–prediction gap and KAPPA (our foil, not our competitor)

Most directly adjacent is work on the knowledge–prediction gap in multiple-choice questions: models fail MCQs whose answers they encode internally, and KAPPA identifies distinct knowledge and prediction subspaces in the residual stream and aligns them with a closed-form, inference-time intervention [29]. We position KAPPA as a foil rather than a competitor, and explicitly *not* as a head-to-head baseline we beat. KAPPA operates where the correct option is present and decodable in the residual (options-present MCQ); our effect lives in cold options-absent text-continuation scoring, where that precondition fails — a faithful k-class knowledge probe in our channel decodes the gold option at chance (best $0.317 \approx$ chance 0.25) while the same probe for the greedy option decodes at 0.542. KAPPA's mechanism therefore does not instantiate in our channel; the relationship is a channel mismatch, and we make no claim of out-performing it. One apparent tension deserves a sentence here, because a reader who reaches Section 5.4 will otherwise trip on it: our read-only selector decodes the gold at **0.943**, far above this 0.317. The two probes read *different channels*. The 0.317 is a k-class probe on the *prompt-final residual* — a single vector with the options in context, which is precisely the channel KAPPA operates on — whereas the 0.943 selector reads the *per-head activations of each option continuation* (four separate forwards, arg-max over the per-option score). KAPPA's precondition fails in our channel, and the selector does not restore it; it succeeds by reading somewhere else. We further reconcile KAPPA's reported free-form *generalization* with our no-transfer result (Section 5.7), since the two superficially conflict: KAPPA's own free-form evaluation carries its gain almost entirely on a single answer-prior-shaped task (TruthfulQA +2.5) while it *regresses* on reasoning generation (GSM8K 91.6→90.7) and is flat on BBQ-Age (+0.2) — so KAPPA, like our lift, does not transfer to reasoning generation, which corroborates rather than contradicts our finding.

2.5 Answer priors and surface-form competition

Our task-heterogeneity finding — that the TruthfulQA half of the lift is largely an answer-prior / surface-form effect — rests on a well-developed line. Surface-form competition shows that highest-probability scoring is biased by how answers are phrased, motivating PMI-style domain-conditional normalization [20]; choices-only probing shows models answer MCQs above chance with the question removed [3, 4]; and choice-sensitivity admits parameter-free fixes [10]. We use the same PMI/decision-conditional and choices-only controls to separate the two tasks: ARC is question-conditional and carries the mechanism, whereas TruthfulQA is largely answer-prior, which is why we report it held-out as evidence the lift is real and never as evidence for the off-axis mechanism.

2.6 The capability tax of interventions, and the specificity question

Adjacent work shows that adapting a model along one axis taxes others: knowledge injection under PEFT erodes reasoning, with the reasoning-critical information distributed across the singular spectrum rather than localized [6], and supervised fine-tuning degrades pretrained capabilities because its targets diverge from the model's autoregressive distribution [25]. Inference-time steering can be applied selectively, choosing when and how much to intervene [18]. These motivate our deployment-pathology results, but they also raise the sharpest reviewer objection to an off-axis claim: that *any* norm-matched perturbation works, so an off-axis direction is not identifiable [34], and even random directions can move behavior [21]. We answer this directly and within the recommended template of comparing a learned direction against a random one [11]: a norm-matched random offset recovers ≈0% of our lift (so the direction is learned, not arbitrary), the strongest on-axis reference recovers 6.4%, and the trained offset has an optimal magnitude (an inverted-U strength sweep), which a generic perturbation would not. We also note that reported specificity failures of activation directions are cross-model; within-model — our setting — directions remain specific [33], so that result does not subsume our control. A related limit on the *write* side sharpens what an off-axis control claim must contend with: a dual-stance evaluation of a sycophancy-reduction direction finds that although sycophantic and factual agreement occupy geometrically *distinct* subspaces, a single steering direction projects equally onto both and cannot differentially target either — a case where representations readable from activations are not separately writable through them [7]. That is the writability gap our read≠write dissociation (Sections 5.4 and 5.5) makes explicit from the opposite side: there a writable direction fails to *separate* two readable targets, whereas here the on-axis correctness direction is readable yet fails to *drive* the lift (W_{know} 6.4%) while an off-axis one does. A distinct mechanistic hypothesis for any logit-level intervention is temperature regulation — confidence-regulation / entropy neurons write into the unembedding null space to rescale the logit temperature without re-ordering the argmax [32] — which we test and rule out for our offset in Section 5.8.

2.7 PEFT, representation editing, and the fine-tuning baseline

Our adapter is from the localized-fine-tuning / representation-intervention family: LoFiT learns additive per-head offset vectors on a sparse set of attention heads [42], a descendant of inference-time intervention [22], and related to representation fine-tuning [40]; concept- and feature-level steering benchmarks find that fine-tuning and prompting often outperform sparse-feature methods [41]. The decisive baseline our inducibility result requires is ordinary supervised fine-tuning: prior domain-adaptation work establishes SFT as a strong, cost-effective default for multiple-choice accuracy while degrading open-ended generation [5]. Our matched Align-LoRA control instantiates exactly this baseline, and our finding — that it recovers ~62% of the contrastive lift — places most of the effect on fine-tuning in general rather than on the contrastive objective.

3 Method

3.1 Model and contrastive-correctness adapter

All results are on Gemma 4 31B IT, run in bf16. The discrimination adapter (referred to as V7-mc) is a contrastive-correctness PEFT in the LoFiT/ITI family: it learns additive per-head offset vectors on a sparse set of attention heads from contrastive pairs of correct and incorrect answers under a multiple-choice discrimination objective. Concretely, each pair is (question, correct option, first-listed incorrect option), built from the TruthfulQA mc1 validation split (seed-0 permutation, 80% train side) and the native ARC-Challenge train split; training minimizes a margin loss on the summed continuation log-probability under the chat template, $-\log p(\text{correct}) + \gamma \cdot \log p(\text{wrong})$ with $\gamma = 0.5$, for 800 steps, with the deployed checkpoint selected by a validation-accuracy monitor. Two construction details a reader of the code will find, stated here rather than left to be discovered. First, because the HF TruthfulQA mc1 release lists the gold answer at index 0, "first-listed incorrect option" means every TQA pair contrasts the dataset's first two options — a fixed positional split rather than a sample over distractors; this touches no mechanism claim (those are anchored on ARC, where option order carries no gold information) and is consistent with — plausibly contributes to — TruthfulQA's answer-prior character (Section 5.9). Second, the checkpoint-selection monitor was, due to dataset ordering in the validation slice, computed on TruthfulQA items only: checkpoint selection never saw ARC signal, so the ARC lift is not checkpoint-selected. The causal analyses use the 48 heads carrying the adapter's offset (selection criterion in Section 3.6). Those 48 (layer, head) targets are unique (no aliasing) and the offset modulates a sparse, *non-contiguous* set of 12 layers — [30, 31, 32, 33, 34, 36, 37, 38, 39, 40, 49, 55] (deterministic across runs; note the gap at layer 35) — so references below to a "layer ~30–55 span" denote this discrete set's bracket, not a contiguous band.

3.2 Evaluation protocol: scoring-face vs generation-face

We distinguish two protocols and report absolutes only with their protocol attached. **Scoring-face** is cold multiple-choice scoring: log-likelihood ranking of options with the chat template applied and no generation (lm-eval canonical, `--apply-chat-template`, n=400, acc_norm), with identical flags across arms. The n=400 sample is the first 400 items of each task's evaluation split in native order (lm-eval `--limit 400`), not a random draw — identical across arms by construction, and not doing selection work: the R4 full-split run confirms every headline delta within the n=400 confidence interval (Section 4.1, Section 7). This is deliberate — it yields apples-to-apples deltas against the adapter — and it is why our base absolutes (e.g. ARC ≈0.50) are scoring-face floors and must not be compared to a third-party-reported ~91% chain-of-thought figure (the protocol caveat in the Table 3 caption). The same scoring-face protocol — identical log-likelihood option ranking, chat template, and lm-eval canonical flags — is also applied out-of-task to GPQA, where the cold-MC lift is itself ~0 (base 0.4747 → adapter 0.4697, −0.5 pp); this cold-MC GPQA measurement is distinct from the generation-face GPQA evaluation described below. **Generation-face** evaluates the produced text: gsm8k under `generate_until` with exact-match; a per-item arbiter that runs base generative chain-of-thought (thinking-on, greedy, k=1) aligned to the cold-MC `doc_ids`, on the items the adapter fixes; and GPQA under thinking-on greedy decoding with the generation budget raised to 8192 tokens (at 2048 the base truncated 96% of generations, producing a spurious 0.15). The per-item arbiter is run across multiple anchors — ARC, GPQA, HellaSwag, and Winogrande — to test whether the fixed items are already CoT-reachable. One caveat is load-bearing here: HellaSwag and Winogrande are not native letter-MCQ tasks, so to run the generative arbiter on them we reformulate each item as a lettered MCQ (HellaSwag: the four continuations → A/B/C/D; Winogrande: the two options → A/B). That reformulation puts the arbiter in a *different* format from the cold-MC scorer (which scores option continuations without letters), so on HellaSwag/Winogrande the arbiter anchors *knowledge* in a letter-MCQ format rather than replicating the cold-MC channel — the same format gap that ARC and GPQA already carry (where the arbiter is nonetheless informative). We therefore read the arbiter as a knowledge-accessibility probe, not as a channel-faithful readout.

3.3 Functional correctness direction and on-axis references

Per attention head we estimate a functional correctness direction v_{inf} (the mass-mean difference between correct and incorrect inference-form activations). The estimation corpus is the inference-family contrastive pairs (ARC-Challenge native train split plus HellaSwag), so on ARC — where every on-axis recovery fraction is anchored — the items that estimate the direction and the test items that measure recovery are disjoint by construction, the same train/test boundary the adapter itself respects (Section 7). To make the on-axis causal test as strong as possible, we also construct W_{know}, a whitened Fisher-LDA correctness direction — the strongest on-axis reference we can build locally, and genuinely distinct from the mass-mean estimate (cos with v_{inf} 0.379). “Strongest” is a construction criterion, not an outcome criterion: the whitened Fisher-LDA direction is the optimal linear correctness discriminant under the class covariance, which the raw mass-mean ignores — it is not chosen for recovering the least lift, and Section 5.5 reports both references’ recovery side by side. The on-axis causal control substitutes an offset built along v_{inf} (and along W_{know}), unit direction scaled to the per-head norm of the trained offset across all 48 heads, so that *direction* is the only variable, gated by a sign-safe three-zone control gate with a null-task check (the gate, and the Gate-A reproduction check it presupposes, are defined in Section 3.6).

3.4 Net-effect geometry and the inducibility control

To compare methods fairly in activation space (rather than via the weight-space offset, which we show is a misleading proxy), we measure each method’s *net effect on activations* and decompose it against v_{inf}, reporting the off-axis/on-axis split as a ratio relative to a random direction. The inducibility control is a non-contrastive Align-LoRA: ordinary supervised fine-tuning on a capacity- and budget-matched corpus, swept over LoRA rank (r8/32/64/256) and scored through the same three-zone control gate, asking what fraction of the contrastive lift generic fine-tuning recovers (this recovery fraction, the budget-matched corpus, and the gate are all defined in Section 3.6).

3.5 Additional controls

We further run: a norm-matched random offset (three seeds) to test whether any perturbation of the right size reproduces the lift; a same-scaffold PMI / decision-conditional decomposition and a choices-only (question-removed) control to separate question-conditional gains from answer priors [20]; layer-band patching (restricting the offset to layer windows) to test for a layer locus; and a within-format difficulty stratification on ARC — ARC-Easy vs ARC-Challenge under the identical cold-MC scoring protocol (acc_norm), stratified by base-wrong items — which holds format and distribution fixed and varies only difficulty, isolating the contribution of difficulty from that of distribution distance.

3.6 Formal definitions and run provenance

This subsection makes the operational terms used throughout precise — each grounded in the analysis code — and maps every headline number to the run that produced it, so no two item populations are silently conflated.

The trained offset θ_{mc}. θ_{mc} denotes the adapter's learned per-head additive offset — the aggregate write direction across the 48 heads (Section 3.1). The geometric and causal decompositions of Section 5.5 ($\mathrm{off}_{\parallel}$/$\mathrm{off}_{\perp}$, whitened cosines) operate on it.

Recovery fraction. For each anchor task the recovery fraction of an arm is $\mathrm{frac} = (\mathrm{arm}-\mathrm{base})/(\mathrm{v7mc}-\mathrm{base})$, the fraction of the contrastive lift the arm reproduces (`scripts/analyze_r1_align_frac.py:6,99`; returns NaN when $|\mathrm{v7mc} - \mathrm{base}| < 10^{-9}$). The per-task headline metric is fixed in code (`TASK_METRIC`, `:36--41`): **acc_norm** for ARC-Challenge and HellaSwag, plain **acc** for Winogrande and TruthfulQA-mc1 (the latter two define no length-normalized variant). Every "fraction of the lift" in this paper is this quantity at these metrics.

Gate-A (canonical reproduction). Before any causal arm is trusted, `scripts/check_canonical_repro.py` re-runs the canonical R1 baselines and aborts unless they reproduce: in *reproduce* mode $|\mathrm{measured} - \mathrm{expected}| \leq 0.05$ for both the base (ARC-Challenge acc_norm 0.505) and the adapter (0.855) (`:69,87--91`); in *null* mode the zero-offset wrapper (α=θ=0) must be behaviourally identical to the base, $|\mathrm{null} - \mathrm{base}| \leq 0.02$ (`:16`), certifying the lift is not a wrapper artifact. A FAIL exits non-zero and blocks the dependent batch. The Gate-A base values are the exact reference the on-axis arms are compared against (e.g. $\mathrm{off}_{\parallel}$ recovering to 0.505 = "0%", Section 5.5).

Three-zone control gate. The inducibility fraction is read through a sign-safe three-zone gate (`scripts/check_canonical_repro.py:70--150`) that keeps numerator and denominator harness-internal (the native HFLM for the control's own effect, the V7 wrapper for the contrastive lift) so no cross-harness bias enters the ratio. Two sign-safety checks precede any fraction: a **setup** gate requires the contrastive lift to be real and positive (v7mc − base > min_effect, min_effect 0.01) and a **convergence** gate requires the control's own effect to be non-trivial and positive (arm − base $\geq$ 0.01), so a no-op or harmful control cannot be scored. The surviving fraction is binned at frac_lo = $1/3$ and frac_hi = $2/3$ into three zones (`:71,73,139--150`); the ~62% ARC recovery lands in the middle zone, and its delta-method fraction interval [45%, 85%] (Section 5.2) is what we report as "zone C" against these thresholds. A separate null-task check (MMLU) guards that a causal arm does not damage an unrelated task (null MMLU +0.07 pp at PASS, Table 9).

Whitened cosine. Geometric alignments (e.g. $\cos(\theta_{\mathrm{mc}}, v_{\mathrm{inf}}) = -0.003$, $\cos(v_{\mathrm{inf}}, v_{\mathrm{ret}}) = +0.283$) are Mahalanobis cosines under the per-head pooled class-activation covariance, in the 256-dim per-head activation space. The canonical headline values come from an eigendecomposition whitening (`scripts/probe_oq1_functional_angle.py::_whitening_from_cov`, reused verbatim by `characterize_theta_mc.py`): $W = V \cdot \mathrm{diag}(\lambda^{-1/2}) \cdot V^{\top}$ from the eigh of cov_common $= \frac{1}{2}$(cov_inf + cov_ret), eigenvalues floor-clamped at 10^{-6} — a floor that never binds in the actual computation (zero clamped eigenvalues; median condition number ~894). Each per-head covariance is estimated from N = 2400 activation samples (1200 correct + 1200 incorrect) against d = 256 dimensions, an N/d ratio of ~9.4. A related additive-ridge formulation, $M = (\mathrm{cov} + 10^{-2} \cdot \mathrm{mean}(\mathrm{diag\,cov}) \cdot I)^{-1}$, appears in a separate probe-direction sanity script and (as a shrinkage on the covariance) in the $\mathrm{off}_{\parallel}$/$\mathrm{off}_{\perp}$ construction; it is not the source of the headline cosines. Robustness: sweeping that ridge factor over two orders of magnitude ($\{10^{-3} \ldots 10^{-1}\}$) moves $\cos(\theta_{\mathrm{mc}}, v_{\mathrm{inf}})$ only within [−0.0032, −0.0021] (always inside the null band, sign stable — stronger regularization pushes it *toward* zero) and $\cos(v_{\mathrm{inf}}, v_{\mathrm{ret}})$ within [+0.284, +0.325] (sign and magnitude stable), so neither headline geometric fact is an artifact of the regularization choice (`runs/oq1_functional_axis/ridge_sensitivity.json`).

The 48 heads. The causal analyses use the 48 (layer, head) cells carrying the adapter's offset. These are the **top-K = 48** heads ranked by per-head probe accuracy (`best_acc` = the max of the Fisher- and Ridge-CV probes), a pure ranking slice with **no accuracy-threshold gate** (`scripts/probe_v7_lofit_head_selection.py:1029--1032`); the 0.60 accuracy threshold in that script only annotates the report, it does not select heads ($K = 48$ follows the LoFiT recommendation).

Align-LoRA corpus and budget. The inducibility control is a non-contrastive LoRA fine-tune under a plain causal-LM cross-entropy loss (`scripts/train_align_lora.py`), on a corpus **matched to the V7-mc training data**: the matched builder (`scripts/build_align_control_corpus.py`) imports only the TruthfulQA and ARC splits used to train V7-mc, byte-identically (TQA = the first 80% of the validation split under a seed-0 permutation; ARC = the native ARC-Challenge train split), and the control is swept over LoRA rank $r \in \{8, 32, 64, 256\}$ at 2 epochs, effective batch 8 (batch 1 × grad-accum 8), max sequence 1024. **"Budget-matched" means the same training examples and the same optimizer steps as the contrastive adapter — not the same FLOPs** (the two objectives differ in per-step cost). ARC is the clean anchor; TruthfulQA is excluded from the fraction because its matched corpus overlaps its evaluation (Section 5.2).

Run → population. The paper's numbers come from a small set of runs with deliberately different item populations; we list them in Table 1 so no two counts are conflated.

Two ARC base-wrong counts appear in this paper — **198** and **204** — and the difference is a **scaffold** difference *inside the same run*, not a difference between runs. Both are the vinf_causal 400. The **lm-eval scaffold** (`d_null.json`, acc_norm, the scoring that produces the headline lift) leaves **198** items wrong; the **premise-file scaffold** (`answer_only_arc_challenge.json`, a length-normalised argmax over option continuations scored under one shared neutral premise) leaves **204**. Every derived subset therefore belongs to one scaffold or the other, and we label them rather than let a reader subtract across the boundary. On the **lm-eval** scaffold: 124 `mc_only`, 151 S_{flip}, 192 $S_{\mathrm{cot\_coldwrong}}$, 202 $S_{\mathrm{base\_right}}$, and **47** adapter-residual (151 + 47 = 198). On the **premise-file** scaffold: 204 base-wrong and **73** adapter-residual, which is the population of the PMI residual-discrimination analysis in Section 5.1. The arbiter and its negative control are reported on the lm-eval scaffold so they are apples-to-apples with the 124, with the premise-file partition given as robustness; both pass an alignment guard on gold index and option count (0 mismatches over 400). The A1 run that supplies the Table 3 headline is a separate run, and its base also leaves 198 ARC items wrong under the same lm-eval scoring (base acc_norm $0.505 \times 400 = 202$ correct).

Run → HF artifact (reproducibility). Several load-bearing numbers come from runs whose result files (small JSON) are synced into `runs/` from a Hugging Face dataset backup, while the heavy binaries they were derived from (per-head cold-MC activations, J-lens jacobians, LoRA adapter weights — each 2–6 GB) remain on HF as pointers rather than versioned in the repo. The pointer for each is given in Table 2.

The post-parser-fix `arbiter_arc_cot_v2.json` (124/124) is versioned in the repo, so `scripts/analyze_arbiter_arc_cot.py` reproduces the headline arbiter count from a clean checkout; the pre-fix `arbiter_arc_cot.json` (123/124, one item lost to a letter-parser artifact) stays HF-only, and the script prefers the v2 when both are present. **Every quantity derived from the arbiter is read from the v2**, and we say so because the correction propagates further than the headline count: two downstream analyses (the cold-MC calibration and the base-arm power check) resolved the arbiter path to the pre-fix v1 by default, which silently excluded doc 128 — base-wrong, and unparsed only because of the very letter-parser artifact the fix removed — from the adapter-independent $S_{\mathrm{cot\_coldwrong}}$ pool. Re-derived on the v2, that pool is **192** rather than 191, its CI lower bound 0.512 rather than 0.513, and the calibration signature is unchanged at the reported precision (z-gap 1.45σ, gold-rank median 2, frac rank≥2 0.59); the base-right control likewise reads 200/202 rather than 196/202. The verdicts are unchanged; we report the re-derivation rather than leave two arbiter versions in circulation. `scripts/analyze_p9_gate.py` recomputes the on-axis-penalty gate from the per-λ files above and self-checks against the reference anchor. The heavy binaries (`vinf_causal/coldmc_perhead_*.pt`, `jspace/{jacobians,gemma_jlens}.pt`, `align_lora_control/*/adapter_model.safetensors`) are HF-only; none of the paper's numbers require re-deriving them.

4 Results

4.1 The headline lift and its non-transfer

The adapter lifts cold multiple-choice scoring substantially (Table 3: ARC +35.0, TruthfulQA mc1 +37.25; transfer benchmarks HellaSwag +17.0, Winogrande +19.0) and loses on generation in every condition (Table 4: gsm8k loss, canonical magnitude **−25** pp per R6 — flat across a generation-budget sweep, so a mechanistic multi-step-reasoning breakage, not truncation). The generation loss that anchors the trade is gsm8k, which is verified and canonical (the R6probe/L3 per-run forensic on the v7mc arm: 295/400 correct, 79 clean-wrong with a normal-terminating wrong `####` numeral, 26 non-terminating reasoning loops, 0 empties) and R6-confirmed; lambada (−32) was an *illustrative* custom-token-norm scorer and the lm-eval confirmation (R9, `lambada_openai` template-off) is **not yet run**, so we do not count it among the verified headline losses (Table 4). The scoring lift and the generation loss are the two halves the rest of the paper reconciles.

The lift is not an artifact of the zero-shot floor, but the anti-floor control resolves into *two* distinct quantities and we report both rather than let one stand in for the other. A paired 5-shot control (pre-registered eval R2, same n=400) raises the base itself — ARC 0.505→0.635, HellaSwag 0.540→0.610, Winogrande 0.6575→0.765 — closing **37% of the ARC headline, 41% of HellaSwag's, and 57% of Winogrande's**. Against that better-prompted base, the **anti-floor** quantity (a@0 − b@5: the zero-shot adapter against a base given every prompting advantage, which is what the kill-rule asks) is ARC **+22.0** pp, HellaSwag **+10.0** pp, Winogrande **+8.25** pp. The **matched-shot** quantity (a@5 − b@5: whether the adapter still wins when few-shot is granted to both arms) is ARC **+22.25** pp (acc_norm, paired McNemar over base-wrong items: 99 fixed vs 10 broken, $p \approx 10^{-19}$), HellaSwag **+17.5** pp, Winogrande **+13.75** pp. On ARC the two nearly coincide (+22.0 vs +22.25) because the adapter is already saturated there (v7mc@5 ≈ v7mc@0); on Winogrande they diverge sharply (+8.25 vs +13.75), and the anti-floor figure is the one we treat as the bound. Per-item on ARC (198 base-wrong at 0-shot), the adapter fixes 151 (76.3%) against few-shot's 69 (34.8%) and covers 64/69 = **92.8**% of the few-shot-fixed items (87 only-adapter, 5 only-few-shot); saturation is task-specific — v7mc@5 ≈ v7mc@0 on ARC (0.8575 vs 0.855) and TruthfulQA (0.81 vs 0.810), while few-shot still stacks on the adapter for HellaSwag (0.785 > 0.710) and Winogrande (0.9025 > 0.8475). The headline remains the zero-shot +35.0, and the anti-floor bound on it is **+22.0**.

The lift is also protocol-bound in a second, sharper sense: with the chat template removed entirely (R1b, same n=400 harness), *both* arms collapse to near-chance on ARC (base acc_norm 0.25, adapter 0.285) — there is no lift to exploit outside the chat-template readout. This is direct evidence that the object of study is the readout protocol itself, and it scopes every claim in this paper to cold-MC scoring under a chat template.

The TruthfulQA +37.25 is reported separately and carries **no** anti-floor control (the lm-eval `truthfulqa_mc1` task fixes num_fewshot=0, so both arms ran 0-shot in R2); only ARC/HellaSwag/Winogrande have a real 5-shot control. We further scope the +37.25 as a **selection lift on the mc1 metric** (paired-significant, Holm-corrected p_{corr} 6.26×10^{-38}), **not** a truthfulness gain and never as evidence for the off-axis mechanism (it is largely answer-prior, Section 5.9).

The held-out-plus-mc2 control has now **run** (full set n=817): the lift **generalizes held-out** (Δ_{UNSEEN} mc1 +37.8 $\approx$ Δ_{SEEN} +36.9; mc2 Δ_{UNSEEN} +26.1), so it is real beyond the mc1 selection metric and is **not** memorization (the matched train covers ~79.9% of the eval, yet the held-out gain is undiminished), whereas a generic Align control collapses held-out (mc1 +7.9), marking the contrastive lift as the genuine, generalizing one. We therefore report it as a real, held-out, generalizing **answer-prior** lift on mc1/mc2 (Section 5.9) — never as truthfulness and never as the off-axis mechanism.

4.2 The fixed items are CoT-reachable — an upper bound on knowledge-accessibility

The non-transfer is not a failure to teach the model anything; it is that the model already knew. Of the 124 ARC items the adapter fixes in cold scoring (and the on-axis offset does not), the base solves 124/124 (1.000) under generative chain-of-thought — the rate of the items it already scores correctly cold (base-right control, 200/202 = 0.990) — so the fixed items are knowledge the base reaches by reasoning but that cold scoring misranks (Table 5). A negative control decides whether that number discriminates or merely restates a ceiling, and it is available from the same run: on the adapter's **residual** — the 47 items that are base-wrong and that the adapter does *not* fix — base CoT accuracy falls to **41/47 = 0.872** (CI $[0.748, 0.940]$), against 151/151 on the base-wrong items it *does* fix (Fisher $p = 1.4 \times 10^{-4}$) and 392/400 = 0.980 over all of ARC ($p = 0.0015$). The arbiter therefore carries information about which items the adapter moves. We attach the magnitude caveat rather than oversell it: ARC sits near its CoT ceiling in every population, so the contrast is statistically clear but absolutely modest, and the residual figure landed between the two thresholds we pre-registered for this control (≤0.85 "discriminates" / ≈0.97 "saturated") — we read it as discriminating on the strength of the contrast, not of the point estimate. We read the 124/124 as an *upper bound* on knowledge-accessibility, not a proof of perfect recall: it is a greedy k=1 arbiter, and for HellaSwag/Winogrande it runs in a letter-MCQ format distinct from the cold-MC channel (Section 3.2). Its robustness is corroborated independently: base self-consistency CoT at k=3 also reaches 124/124 on this set (Section 8), so the result is not an artifact of the single greedy chain. This holds out-of-task too: on GPQA the base solves the items under chain-of-thought at 0.7071 (arbiter), well above chance, so there is no generative headroom a scoring shortcut could supply, and we therefore did not build a verifier-guided arm (a design choice following from the no-headroom result, not an independent test). And measured directly, the cold-MC lift on GPQA is itself ~0 (base 0.4747 → adapter 0.4697, −0.5 pp, CI crossing zero): the no-transfer is confirmed out-of-task (GPQA is out-of-task and distribution-distant — it is also the B.2 spillover task at −6.6 pp — so this is a no-transfer-out-of-task result, not a no-transfer-to-hard one; Section 5.1 separates difficulty from distribution with within-format and second-anchor controls). The adapter is not inert there — it fixes ~21 of 104 base-wrong items at a 0.202 fix-rate and breaks ~22 base-right ones for a net of about −1 — but that exchange is churn within noise (no McNemar test), not compensatory damage. Note the converse implication we do not hide: GPQA leaves ~23 pp of cold-scoring headroom (cold 0.4747 vs CoT 0.7071) that the adapter fails to recover — the adapter is not a general repair of the lossy readout but an in-distribution exploit of it, which is exactly what the off-axis-of-degree geometry predicts.

4.3 Most of the lift is generic to fine-tuning

A matched non-contrastive Align-LoRA recovers ~62% of the ARC lift, flat across a rank sweep (Table 7: 59/63/60/65; slope-CI $[-1.4, +3.4]$ includes zero, so the residual is not a closeable capacity gap), placing most of the effect on fine-tuning in general rather than on the contrastive objective; the contrastive-specific remainder is at most ~38% and is an upper bound: a best-effort fairness control **ran** and does **not** raise the

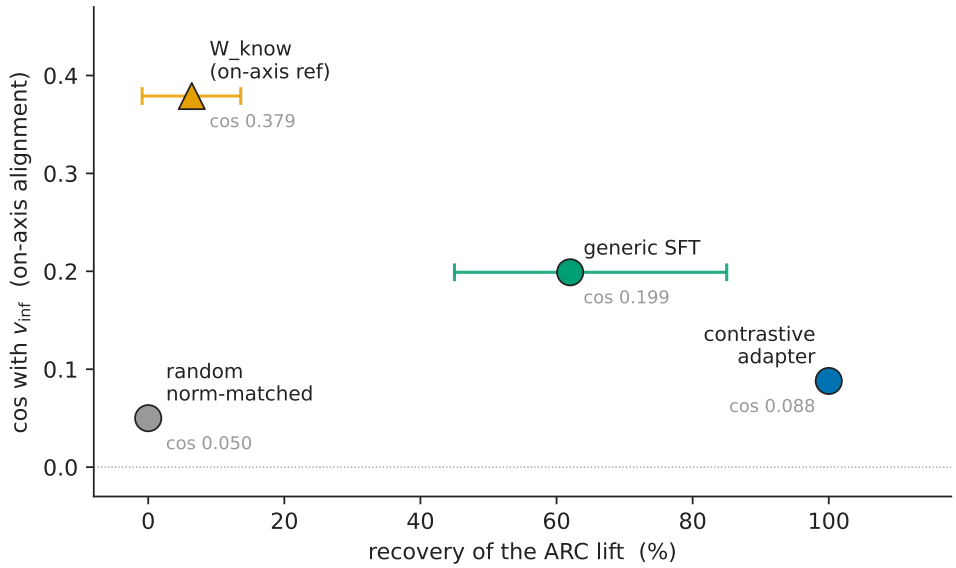

Figure 1: The matched-control spectrum on two independent axes. x: fraction of the ARC lift each direction recovers; y: cosine with the functional correctness direction v_{inf} (net effect on activations for the three methods; the reference direction itself for W_{know}, marked ▲). The two orderings are not monotone in each other: the most on-axis point (W_{know}, cos 0.379) recovers only 6.4%, while the top recoverer (the contrastive adapter, 100%) is near-off-axis (cos 0.088). Most of the lift is generic to fine-tuning (generic SFT ~62%), and where the recovery lives is a method-dependent degree of off-axis, not a single fact. Horizontal bars are the published 95% intervals on the recovery fraction where one exists — W_{know} [−0.9%, +13.6%] and generic SFT [45%, 85%]; the random floor (three seeds) and the contrastive anchor (100% by construction) carry none, and we do not synthesise one. **W_{know}'s interval crosses zero and overlaps the random floor:** its 6.4% is statistically indistinguishable from no recovery (exact McNemar $p = 0.136$), so the spectrum is three rungs — floor → generic SFT → contrastive — not four. The generic-SFT interval is correspondingly wide, which is why the claim it supports is *most of the lift*, not a specific percentage. Values in Table 8.

~62% — increasing rank to r512 recovers *less* (frac 0.22 at lr1e-4, converged but overfitting) or yields a below-base broken adapter (lr2e-4), so ~62% is a **tight** lower bound and the ~38% remainder a firm upper bound. TruthfulQA is excluded from this control as train-on-eval leakage (held-out, the highest-rank control collapses 98%→30%).

5 Discussion: the mechanism of the recovery

The scoring-face gain documented above invites a natural reading: the contrastive-correctness adapter sharpens the model's internal direction for answer correctness, so that cold multiple-choice scoring becomes more accurate. This section shows the gain is better understood as a *quantified causal attribution* than as a sharpening. We take as established setup that cold multiple-choice scoring under a chat template under-expresses knowledge the model holds (Section 2; 11, 15, 29); the result is what the adapter does about it. Most of the recovery is what *generic* fine-tuning buys (Section 5.2); it lives in a learned direction that is off-axis-of-degree to the model's correctness geometry and is causally privileged within that off-axis complement (Section 5.5); it has no layer locus (Section 5.6); and it does not transfer to generation (Section 5.7).

The off-axis geometry is not a curiosity but the *expected* signature of the attribution we lead with, and reading it that way organizes the rest of this section. The cold readout does not fail for lack of knowledge; it fails by **surface-form swamping**: on the items the adapter fixes the base buries the gold beneath more fluent distractors (Section 5.1). Plausibly, though we do not measure this, suppressing surface form need not align with signalling correctness. The two causal arms make the point in opposite directions: injecting correctness *on-axis* (the strongest on-axis reference we can build, W_{know}) recovers only **6.4%** of the lift, which refutes a knowledge-deficit account; pushing *off-axis* (perpendicular to correctness, $\text{off}_{\perp}$) recovers ~**100%**, which is consistent with the surface-swamping account. The off-axis direction is therefore not a spurious accident but a natural form for a fix to take when the pathology is one of format rather than of knowledge — which is why a single off-axis push unifies the readout result (Section 5.1), the inducibility attribution (Section 5.2), the on-axis null (Section 5.5), and the deployment pathologies (Section 5.7).

5.1 The fixed items are chain-of-thought-accessible, the no-transfer is out-of-task (not difficulty-bound), and the readout misranks by confident burial

The cleanest evidence that the readout — not the model's knowledge — is the bottleneck comes from a per-item arbiter. We isolate the **124** ARC items the adapter corrects in cold multiple-choice scoring but the on-axis offset does not (Section 5.5), and for each ask whether the *base* model already reaches the right answer through generative chain-of-thought. It does: the base solves **124/124** (1.000) of these items under generative CoT (the fixed-parser re-run resolves the earlier 123/124: the one "failure" was a parser artifact — doc 128's numeric option label "4", which the letter-parser could not emit by construction — now parsing correct, n_unparsed 5⟶0), indistinguishable from its CoT accuracy on the items it already scores correctly cold (base-right control, 200/202 = 0.990 — read from the same fixed-parser run, not from the superseded pre-fix one, where it was 196/202). The items the adapter "fixes" are knowledge the base already reaches by reasoning; cold scoring simply misranks them.

The arbiter discriminates: a negative control on the adapter's residual. A per-item arbiter of this shape invites an obvious objection — if the base solved *everything* under chain-of-thought, "the fixed items were already CoT-accessible" would be true of all of ARC and would say nothing about the adapter. The control is the complementary population, and it needs no new generation: the v2 arbiter run already covers all 400 items under the identical protocol (greedy k=1, same prompt, same fixed parser, `n_unparsed=0`), so re-partitioning it is protocol-identical to the headline by construction. On the adapter's **residual** — the 47 base-wrong items it does not fix — the base solves $\mathbf{41/47 = 0.872}$ (CI $[0.748, 0.940]$), against $\mathbf{151/151}$ on the base-wrong items it does fix (Fisher $p = 1.4 \times 10^{-4}$) and $\mathbf{392/400 = 0.980}$ over all of ARC ($p = 0.0015$). A related partition in the independent premise-file scaffold — which defines the n=73 residual of the PMI residual-discrimination analysis below; the calibration analysis runs on the lm-eval scaffold — agrees: $66/73 = 0.904$ against $326/327 = 0.997$, $p = 3.6 \times 10^{-5}$. It is *related* rather than identical, and we say so: the lm-eval partition conditions on base-wrong (47 against 151), while the n=73 is every item the adapter gets wrong, including 8 it broke from base-right. Restricting it to base-wrong for a like-for-like read gives 58/65 = 0.892, between the two. So the 124/124 is not trivially true of all of ARC. Two honest qualifications travel with it. First, the magnitude is modest in absolute terms — ARC is near its CoT ceiling everywhere, and the discrimination is 0.872 against 0.980, not against something far lower. Second, the residual figure fell *between* the two thresholds we pre-registered for this control ($\leq$0.85 → discriminates, $\approx$0.97 → saturated); we therefore rest the reading on the contrast and its interval (CI upper 0.940 excludes the saturated value) rather than on the point estimate clearing a bar it did not clear. The adapter is a *shortcut* to that knowledge — it nudges the scoring logits toward answers the base could reason its way to, without reasoning.

This directly explains why the lift does not transfer to generation (Section 4): on exactly these items the base is already at ceiling under CoT, so there is no generative headroom for a scoring shortcut to add. The obvious objection is that ARC is simply easy — near its CoT ceiling — so the arbiter cannot tell "recovers knowable" from "fabricates plausible." We corroborate the no-transfer out-of-task.

On **GPQA**, where the base must genuinely reason, its CoT accuracy is **0.7071** (arbiter; 140/198, thinking-on, with the generation budget raised to 8192 tokens — at 2048 the base truncated 96% of its reasoning, producing a spurious 0.15). Two things follow. First, 0.7071 is well above chance, so the base *does* solve these by reasoning; yet the cold-scoring shortcut still has no generative headroom to add, and indeed the measured cold-MC lift on GPQA is ~0 (base 0.4747 → adapter 0.4697, −0.5 pp, CI crossing zero) — the no-transfer result restated out-of-task. This is a no-transfer-*out-of-task* result, not a no-transfer-to-*hard* one: GPQA is out-of-task and distribution-distant (it is the B.2 spillover task at −6.6 pp). Second, because the base already reasons GPQA out, a verifier-guided arm (base generates, adapter scores) has nothing to harvest, so we did not build one (a design choice following from the result, not an independent test). The readout-is-the-bottleneck result therefore survives an out-of-task control that the ARC arbiter alone could not provide.

Two added controls now separate difficulty from distribution, where previously a single out-of-task benchmark could not. *Within-format difficulty:* the cold-MC fix-rate on base-wrong items moves only **−0.10** as ARC difficulty rises (ARC-Easy 0.861, n=360 → ARC-Challenge 0.763, n=198, both burial-matched by gold-rank), whereas it collapses **−0.56** going out-of-task to GPQA (0.202) — so *distribution*, not within-format difficulty, drives the collapse. *A second pair of anchors:* on HellaSwag and Winogrande, where the lift also holds (+17 / +19), the base's CoT accuracy is high (0.83 / 0.9425) — like ARC's 0.97 and well above GPQA's 0.70 — and the recovery is recoverability-gated (the adapter fixes CoT-right items more than CoT-wrong ones: HellaSwag 0.64 vs 0.31, Winogrande 0.76 vs 0.42), i.e. the lift rescues knowledge the base can already reason out. The no-transfer is therefore **out-of-task, not difficulty-bound** — supported now, not merely suggestive, on all three anchors. *(Power caveat: the hard cells are small — HellaSwag n=32, Winogrande n=12 — and the Easy→Challenge difficulty range stays below GPQA's, both base-CoT near 0.97; the burial gradient is shallow throughout, corr(fix, z-gap) −0.17.)*

How the readout misranks on ARC: confident burial under fluency. The arbiter shows the model *has* the answer; a calibration analysis on ARC shows *how* cold scoring loses it, and it is not indifference between close options but misplaced confidence. We measure this on an **adapter-independent** subset — the n=192 ARC items the base scores wrong cold yet solves under CoT ($S_{\text{cot\_coldwrong}}$, defined without any reference to the adapter) — so the signature cannot be an artifact of which items the adapter happened to fix. On these items the base's chosen (wrong) option beats the gold by **1.45σ** within-item, the gold sits at **median rank 2 of 4** (only 41% have the gold at rank 1), and a gold-vs-distractor ranking on the base-wrong items scores **AUC 0.394, *below* chance**. (This confident-wrong signature is the multiple-choice overconfidence aligned-model calibration work documents, which separates an *answer-decision* uncertainty from a *format-preference* uncertainty and attributes alignment overconfidence to their conflation [17]; our claim goes past calibration — the per-item arbiter shows the buried answer is *reachable*, at 124/124 under CoT, and a learned offset *exploits* the gap rather than re-calibrating it post hoc.) The gold is not a flat near-miss; it is actively *buried* beneath more fluent surface forms — the base's raw pick is the gold only **19%** of the time here, against **81%** on the items it scores correctly cold.

The re-ranking the adapter performs is **promotion-dominated** (it adds 0.35 of probability mass to the gold while removing 0.21 from wrong options, and sharpens the distribution, Δentropy −0.19), consistent with surfacing a buried-but-present signal rather than manufacturing one.

Methodological note we are forced to make: the apparent *flatness* of the char-normalized answer entropy (0.920 on this subset) is a softmax-temperature artifact — length normalization compresses the spread ~50× without reordering, while the raw distribution is peaked. We therefore report **rank and within-item z-gap**, which are invariant to that normalization, and not entropy, whose flatness flips with the metric. (The selection-conditioned S_{flip} subset, the 151 items the adapter flips, shows the identical signature — z-gap 1.37σ, gold-rank median 2 — and serves only as a consistency check; its conditioning on the adapter's flips is why the adapter-independent subset is the headline.)

This burial-and-recoverability signature is established on ARC; we **could not detect** it out-of-task on GPQA, but as a power-starved non-finding rather than a contrast: on the burial-matched hard cell the CoT-right and CoT-wrong fix-rates are equal (0.186 = 0.186), but n=43 with 31 items left unparsed (k=1 greedy decoding) contaminates that cell, leaving it indistinguishable from no measurement. We therefore report the recoverability-gating as **could-not-detect / power-starved out-of-task**, not as evidence against it.

The signal is swamped, not absent — "broken" means *lossy*. It does not follow that the cold channel is empty. After removing length and fluency (a PMI / decision-conditional residual), a gold-vs-distractor discrimination scores above chance on both arms in the *same scalar channel*: **AUC 0.574** (CI $[0.520, 0.627]$) on the base-wrong items (n=204), and **0.667** (CI $[0.584, 0.744]$) on the adapter's own residual (n=73, the items it does not fix). The correctness signal is present in the cold channel but *out-ranked* by fluency, not erased: the same PMI-residual channel carries gold-vs-distractor signal, rather than the channel being empty and the adapter supplying missing knowledge.

Honesty boundary: these are two arm readings on **different populations** (n=204 vs n=73), not a within-set rise; and the base-arm CI lower bound, 0.520, does not on its own clear a 0.55 "present" threshold — we call the base channel *intermediate* and rest the present-but-swamped claim on the adapter arm (CI lower bound 0.584 > 0.55), and raising power on the base arm did not rescue it: pooling base-wrong with $S_{\text{cot\_coldwrong}}$ — and every defensible pool variant — ran, and none clears 0.55. The adapter-independent $S_{\text{cot\_coldwrong}}$ (n=192, the items the base *provably* knows via CoT yet cold-MC misranks) reaches only CI lower bound 0.512, and even the literal union base-wrong $\cup$ $S_{\text{cot\_coldwrong}}$ (n=228, which is *pro*-gate — its 24 extra items are base-right under premise scoring) reaches 0.535, so the base channel stays *intermediate* on the strongest honest pool.

5.2 The recovery is mostly generic to fine-tuning, not special to the contrastive objective

If the lift were a specific consequence of the *contrastive-correctness* objective, an ordinary supervised fine-tune of matched capacity and budget should not reproduce it. It largely does. We train a non-contrastive Align-LoRA on the same data budget and run it through the same causal recovery machinery (a sign-safe three-zone control gate). It recovers about **62%** of the ARC lift, and that fraction is **flat across a rank sweep** (r8/32/64/256: 59/63/60/65), with the sweep slope's confidence interval spanning zero ($[-1.4, +3.4]$) and a delta-method fraction interval of roughly $[45\%, 85\%]$ (zone C). The residual is therefore not a capacity gap the generic control can close with more parameters; ordinary fine-tuning simply buys back most of what cold scoring leaves on the table. One caveat we apply to ourselves, symmetric with the information-set discipline of Section 5.3: "matched capacity and budget" matches examples and optimizer steps, not the effective training signal per step — the contrastive objective sees each example with an explicit correct/incorrect contrast where the SFT objective sees only the correct completion — so the ~62% attributes the lift to *fine-tuning on this data* rather than to any stronger claim of per-step informational equivalence between the two objectives. The contrastive-specific remainder is at most ~38% of the ARC lift, and this is a **firm upper bound**: a best-effort fairness control has now run and does **not** raise the generic fraction — raising the rank to r512 recovers *less* (frac 0.22 at lr1e-4, converged but overfitting ep2→ep4, robust ep2=ep4) or yields a below-base broken adapter (lr2e-4, excluded like the r256hi high-LR control), so the ~62% is the ceiling of generic-fine-tuning capacity, a **tight** lower bound. That de-risk (higher rank, an LR sweep, more epochs) is now closed; it refined the residual, not the thesis.

We restrict this control to ARC because TruthfulQA's matched corpus leaks into its evaluation. Across the rank sweep the TQA recovered fraction rises monotonically (41/66/72/91), the signature of memorizing training examples that recur at test; held out on 20% of TQA, the highest-rank control collapses from 98% to 30% recovery and the rank sweep flattens (r8 = r256 = 0.524), while the contrastive adapter still generalizes held-out (0.83). The matched corpus (TQA train-80%, seed 0) overlaps the evaluation set; TQA is therefore excluded from the inducibility control, and the ~62% generic figure is an ARC result.

5.3 The lift does not reduce to an interpretable re-scoring of the output scalars

The inducibility result attributes most of the lift to generic fine-tuning, but it leaves a sharper question: whether the recovered signal is a simple, interpretable re-weighting of the scores the cold readout already produces (length, PMI, an answer prior, a calibration temperature). If so, the "lost" knowledge would be recoverable by post-hoc re-scoring alone, without touching the model's activations. It is not. We build the strongest such re-scorer we can — a supervised, held-out, AUC-objective combination of the interpretable surface features (conditional log-likelihood, answer-only log-likelihood, character length, PMI) — and ask how well it separates gold from distractor on the base-wrong items. It caps at **AUC 0.574** (a held-out logistic fit at 0.458 and a non-linear GBM at 0.462, an on-test-tuned oracle stack at 0.502, PMI alone at 0.574), against the **adapter's 0.830** on the same items. PMI does not merely fail to match the adapter — applied *to* the adapter it *hurts* it (0.818 → 0.758), and the adapter's re-ranking is only **2.4%** explained by the surface features (R^2 0.024). The recovered signal therefore lives **upstream of the collapsed output scalars**: no re-scoring of those scalars recovers the lift, which answers the "your baseline is just under-fit" objection — the ceiling is supervised and held-out, not a hand-tuned stack.

We are deliberate about what this licenses. The comparison is *information-set-confounded* — the adapter mutates **activations**, whereas the surface stack re-weights **collapsed output scalars** — so the gap (0.830 vs 0.574) does **not** license "a magic off-axis direction" (a generic SFT recovers ~62% the same way, Section 5.2) nor "an empty channel" (the residual is 0.574 > chance, Section 5.1). The defensible claim is about the **channel**: the lossy readout discards signal that survives in the activations and is not reconstructible from its own outputs.

We can now sharpen the channel claim with a same-space probe. A logistic probe fit on the *same* per-head `o_proj`-input activations the adapter writes to — no fine-tuning, held out by item, on the same cold-MC option-continuation distribution, scaler→PCA→logistic with the preprocessing fit inside each fold and a within-item label-permutation null — recovers gold-vs-distractor on ARC base-wrong items at **AUC 0.955** (CI [0.934, 0.973], permutation null 0.494), far above the **0.502** collapsed-output ceiling. A surface-form control — the same probe on length/format features of the option string — caps at **0.442** (≈ chance), so the 0.955 reads correctness the output scalars discard, not a length/format artifact. This **widens and cleans the channel gap** (0.502→0.955) relative to the scalar-stack version (0.574→0.830).

We are careful about what it does **not** license. First, the probe reads 12288 activation dimensions while the adapter is constrained to nudge output logits. The two are not on the same axis, so 0.955 is **not** "the probe beats the adapter," and we do not read it as a representational *edit*. Second, that a linear probe decodes correctness from activations at ~0.95 is a near-universal property of capable language models [2, 26]. It shows the information *exists* in the activations the readout collapses, **not** that the model's own deployed readout is "merely miscalibrated." We therefore make a **sharpened channel claim** — the cold-MC output-scalar channel is lossy relative to the activation channel — and we do **not** assert "pure miscalibration" or "signal intact upstream" as anything beyond that channel statement.

Third, the recoverable signal is **broadly distributed**: a random 48-head set decodes it as well (0.949 ≈ the adapter's 48 heads), so the probe speaks to the cold-MC channel being lossy *model-wide*, not to the adapter's subspace being privileged. This is a read-test about the channel, decoupled from the write-tests that establish the adapter's specificity (a random norm-matched *offset* recovers ~0% of the *lift*, Section 5.5, a different operation). On TruthfulQA the same probe reaches 0.974, but its surface-form control is non-trivial (0.639, with a gold-first position artifact at 1.000), consistent with TruthfulQA's answer-prior character (Section 5.9); the channel claim is therefore anchored on ARC.

5.4 A read-only selector deploys the recovered signal without the pathologies

The channel result has a constructive consequence — but we scope it as the *read leg of a read≠write dissociation*, not as a new method, because the bare capability is already published. That a read-only probe or readout can *select* multiple-choice answers and recover an accuracy lift, with no weight edit and no generation, is established by at least two independent mechanisms: NoVo selects answers by voting over truth-correlated attention-head norms (+19 points on TruthfulQA-MC1, gains on >90% of 20 datasets) [19], and CORAL re-scores per-option probabilities with a frozen weight-decay correctness probe (+14% accuracy on ARC-Challenge/HellaSwag/OpenBookQA/Math-MC, 7B models) [27]. Both present the selector as a calibration/accuracy *fix*; our use is narrower and different — the read-only selector is the *evidence* that the lift the adapter writes off-axis is recoverable read-only, on-axis, without the write's damage (a read≠write dissociation neither studies, lacking an off-axis adapter to audit). If the correctness signal the cold readout discards is *present and read-only recoverable* in the activations, then one can recover the MC lift without writing any offset at all — and therefore without the deployment pathologies the write incurs (Section 5.7).

We deploy the same-space probe of Section 5.3 as a **read-only selector**: for each cold-MC item it reads the per-head activations and picks the option the probe scores highest, with no offset injected and no change to generation. On ARC it reaches selector accuracy **0.943** (± 0.002 over three seeds), against the base's 0.505 and the adapter's 0.853, and the gain is not a correct-for-wrong trade: relative to the base it fixes **181** of the 198 base-wrong items and breaks only **5** of the 202 base-right ones (exact McNemar $p = 3.7 \times 10^{-47}$; an accuracy-space permutation null sits at 0.255). Because it never touches the weights or the decoder, it carries none of the adapter's costs — no retrieval spillover, no generation loss (no reasoning breakage) — so on the (cost, MC-accuracy) plane it **Pareto-dominates the adapter**. Whether it also wins on deployment grounds — against the *reasoning* baseline the no-headroom arbiter (Section 5.1, base 124/124 under CoT) suggests, and against the trivial one-forward baseline the review asked for — is settled below by the deployment triangle: it does **not**. A reasoning TTC ties the cheapest baseline, and a single-forward letter-emission of the base dominates the selector itself at equal cost — which is exactly why we hold the selector to its *evidentiary* role (read≠write) rather than a deployment win.

We hold the claim to its read side and guard against an over-reading: 0.943 exceeding the adapter's 0.853 is **not** "the probe beats the adapter as a capability" — the probe reads 12288 activation dimensions while the adapter is constrained to nudge collapsed output logits, so the two are not on the same axis (Section 5.3). The claim is that the cold-MC channel *discards* a signal a linear reader recovers read-only. One caveat bounds the contribution to Tier-1: the 48-cell feature set is held out by *item* but selected on the training split, not held out by *feature*, so the decisive strengthening is cross-benchmark (ARC-Easy / OpenBookQA), which we flag as the immediate follow-up.

The deployment triangle (Figure 2). The read-only selector is neither the strongest nor the cheapest good way to answer these items — both distinctions belong to the base model asked correctly. On the 198 base-wrong ARC items, four deployment routes separate on the (accuracy, cost) plane. Three cost a single forward: the contrastive adapter's cold-MC scoring recovers **0.763**; the read-only selector recovers **0.914**; and — the same-cost baseline the review asked us to run — prompting the base to emit the answer letter in one greedy forward, no chain of thought, recovers **0.970** (192/198; 2 items unparsed and counted wrong, so a floor). The fourth, a reasoning test-time-compute baseline (base self-consistency CoT, k=3), also recovers **0.970**, but at ~1179 generated tokens. So the single cheapest intervention — one forward, letter emission, no adapter and no probe — matches the expensive reasoning baseline and dominates both the adapter and the selector at equal cost.

One caveat scopes this exactly as it scopes the arbiter (Section 3.2): letter emission puts the base in a lettered-MCQ format, a *different readout* from the cold-MC option-continuation scoring the adapter operates

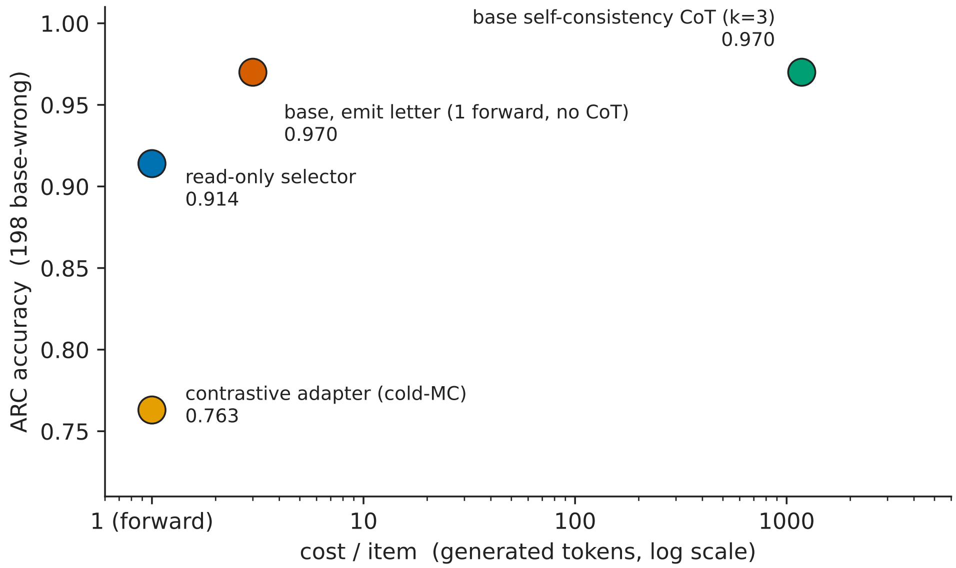

Figure 2: The deployment triangle on the 198 base-wrong ARC items. x: cost per item in generated tokens (log scale; a single forward is one unit); y: ARC accuracy. Four routes: the contrastive adapter's cold-MC scoring (0.763, one forward) is dominated by everything; the read-only selector (0.914, one forward) improves on it read-only; **the base asked to emit the answer letter in one forward (0.970) matches the reasoning baseline at** ~1000× **less cost and dominates both the adapter and the selector at equal cost**; base self-consistency CoT (0.970, ~1179 tokens) ties letter-emission at far higher cost. The whole LoFiT/ITI-for-cold-MC family optimizes a readout that the most trivial same-cost alternative — ask the base for the letter — already beats. **The comparison is evidentiary, not a like-for-like scoring test:** letter emission is a different (lettered-MCQ) readout than the cold-MC channel (Section 3.2), so the triangle contrasts two readouts of the same base at equal cost — the lossy-readout thesis in its sharpest form — and the selector's role stays evidentiary (read≠write), not a deployment recommendation, since even it is dominated by letter emission.

on — not a like-for-like scoring comparison. That is the point rather than a confound: it compares two readouts of the *same base* at the *same cost* — loglik continuation ranking vs letter emission — which move it from 0.505 to 0.980 over the full set (0→0.970 on the base-wrong subset the triangle uses). This is the paper's thesis, that the cold-MC channel is lossy, in its sharpest form: the knowledge is present and cheaply reachable by a different readout of the base, and the adapter's off-axis write recovers *less* of it (0.763) than the most trivial same-cost alternative (0.970).

The selector Pareto-dominates only the *adapter*; a reasoning TTC and a one-forward letter-emission of the base both beat it — so the whole LoFiT/ITI-for-cold-MC family optimizes a readout that even asking the base for the letter already beats. We therefore report the selector as *evidentiary* (the correctness signal is read-only recoverable, read≠write) and **not** as a deployment win (dominated at equal cost by letter emission). *Population note (do not conflate):* the four figures here (letter-emission 0.970, SC-CoT 0.970, selector 0.914, adapter 0.763) are all on the **198 base-wrong** subset; the selector's all-items accuracy (**0.943**, three seeds, across all n=400 evaluation items) and the adapter's all-items accuracy (**0.853/0.855**; Table 3) are computed on a *different* population and are not directly comparable to the 0.914 / 0.763 here. The 0.943 and 0.914 are the same selector on two populations, not two systems. "All items" here means the n=400 harness sample, **not** the full ARC-Challenge test split — that phrase is reserved for the R4 full-split run (Section 7, ARC acc_norm +36.2).

read ≠ write, geometrically. The selector and the off-axis write together make a four-way dissociation the paper rests on. A reader *reads* correctness from the activations (selector 0.943); an *on-axis* write fails to recover the lift (W_{know} 6.4%); an *off-axis* write succeeds (off$_\perp$ ~100%, shuffle ~0%, Section 5.5); and — closing the loop — the probe's decision direction is geometrically aligned with the correctness axis it reads, *not* with the axis the adapter writes. Per head, the probe direction's signed cosine with the functional correctness direction v_{inf} is +0.22 (median +0.20, above the ~0.06 null band), while its cosine with the adapter's write direction θ_{mc} is |0.066| (at the edge of the ~0.061 per-head null band, not clearly within); pooled, +0.209 vs |0.008| (clearly within the null — the read≠write conclusion rests on the pooled statistic). The read axis is v_{inf}; the write axis is off-axis to it. We mark this geometric leg **moderate, not a slam-dunk**: the alignment is 0.22 rather than 0.5, fold-stability is 0.887 (marginal at $n \ll d = 12288$), and v_{inf} lives strongly on the top principal component (cos 0.55) — but the probe reads v_{inf} 66% *beyond* PC1 (its PC1-routed contribution, $0.134 \times 0.55 = 0.074$, is well under the 0.219 total), so the alignment is a genuine read, not a by-product of dominant variance. The read ≠ write conclusion does not hinge on this leg — it is already carried by the probe (read), the W_{know} null (on-axis write fails), and the off$_\perp$/shuffle pair (off-axis write succeeds) — but the geometry confirms it directionally.

5.5 The recovery is off-axis to correctness — of degree, on a measured spectrum

We now locate the recovery relative to the model's own correctness geometry. Per attention head, we estimate a functional correctness direction v_{inf} and compare it both to the adapter's trained offset and, more carefully, to the *net effect each method has on activations*. The trained offset alone is a misleading proxy: its whitened cosine with v_{inf} (Section 3.6) is −0.003 (within the random-direction null), it places only 6.4% of its mass in the top principal components (a quantity unrelated to — and only coincidentally equal to — the W_{know} recovery figure below), and its norm is 4.3× the class gap — a near-orthogonal weight-space picture. But the offset is a weight-space proxy, and a zero-cost preview of the net-effect comparison from weights alone reported "off-axis" (ratio ~1.05); the clean experiment — comparing the net effect on activations — refutes that simplicity.

The clean comparison yields a **spectrum**, not a single off-axis fact. A norm-matched **random** direction (three seeds) recovers ≈0% of the lift (ARC acc_norm 0.50–0.52 ≈ base 0.505): the learned direction is specific, not any perturbation of the right size. The strongest **on-axis** reference we can construct locally — a whitened Fisher-LDA direction W_{know}, genuinely distinct from the mass-mean (cos with v_{inf} 0.379) yet orthogonal to the offset (cos with θ_{mc} 0.0014) — recovers only **6.4**% of the ARC lift (null 0.500 / adapter 0.853 / W_{know} 0.522; fraction 0.064, far below the pre-registered 0.30 kill-rule and statistically indistinguishable from zero — exact McNemar $p = 0.136$, with the paired-bootstrap recovery CI $[-0.009, +0.136]$ and the Wilson accuracy CI $[0.474, 0.571]$ both including chance, which *strengthens* the on-axis-fails reading), and essentially nothing on TruthfulQA; a raw mass-mean v_{inf} recovers 11% (+4.0 vs +35.0; $z = -6.63$). The ordering of the two on-axis references deserves a sentence, because a reader may suspect the smaller number was chosen for convenience: the better-constructed discriminator (W_{know}, the covariance-optimal linear separator, Section 3.3) recovers no *more* of the lift than the plain mass-mean (6.4% vs 11%) — improving the on-axis estimate does not improve recovery, which is itself evidence against a knowledge-deficit account. Both figures sit far below the 0.30 kill-rule, and the conclusion is unchanged whichever is taken as the on-axis ceiling.

This off-axis picture does not hinge on the on-axis references being weak estimators of the true correctness axis: the offset is off-axis **even to the probe's own decision direction** — the empirically optimal correctness reader — the in-space probe of Section 5.3, whose base-wrong AUC is 0.955 (distinct from the Section 5.4 selector's accuracy, 0.943 at n=400 / 0.914 on base-wrong) — at pooled cosine |0.008|, inside the null band (the per-head |0.066| sits at the edge of the ~0.061 per-head null, so this upgrade rests on the pooled statistic).

The two *methods*, measured by net activation effect, are both **mostly off-axis** — the contrastive adapter at cosine 0.088 (ratio 1.74 vs a random direction) and generic SFT at cosine 0.199 (ratio 3.93) — but generic SFT is **2.3× more on-axis** than the contrastive adapter. The recovery is thus off the correctness axis *to a degree that depends on the method*: random ≈0% (cos≈0.05, ratio 1.0) → W_{know} 6.4% (the on-axis reference) → generic SFT ~62% (cos 0.199, ratio 3.93) → contrastive 100% (cos 0.088, ratio 1.74). Note the two orderings are deliberately **not** monotone in each other: recovery fraction and on-axis-ness are different axes of the spectrum (the most off-axis direction, random, recovers nothing; the more on-axis method, generic SFT, recovers *less* than the contrastive one) — the spectrum is a two-axis object, not a single scale.

This refines the earlier "purely off-axis" framing into the central figure of the paper, and it is consistent with the earlier finding that the two functional correctness faces are themselves co-located (whitened cosine $v_{\text{inf}} \cdot v_{\text{ret}}$ +0.283, BCa95 $[+0.276, +0.291]$, null ±0.007, n=316 heads): the recovery is off the shared correctness substrate by a method-dependent amount, not aligned with either face of it.

This co-location appears to sit in tension with work reporting that recall and reasoning occupy *causally separable* circuits [14]: if the correctness faces share a substrate, separately ablating recall and reasoning should not be possible. The tension dissolves at the level of description, because the two results measure different objects. That work measures task-circuit separability (in Qwen and LLaMA): ablating one functional circuit leaves the other intact, "separable but interacting." We measure the *direction of a trained offset* relative to the functional correctness direction, in Gemma 4 31B IT. A substrate can host circuits that partition sufficiently for separate ablation *and* carry functional directions that are co-located, with our +0.283 measurement living in that work's "interacting" seam rather than denying its separability. Our novel layer is neither: the direction the adapter actually moves is off-axis to both functional faces, which is why a partition-based account of the spillover — and the hope of re-selecting heads to avoid it — does not hold.

A direct causal decomposition: the off-axis push is privileged, not any perturbation of the right size. The spectrum above is read from each method's net activation effect; a second, more direct experiment decomposes the adapter's *own* offset against the correctness direction and asks which component carries the lift. We split the trained offset, per head, into a component parallel to the functional correctness direction ($\text{off}_{\parallel}$) and a component perpendicular to it ($\text{off}_{\perp}$), and inject each through the identical recovery harness used for every arm above, at its natural norm from the decomposition ($\text{off}_{\parallel}$ is consequently ~16× smaller in norm than the W_{know} arm — the equal-norm control inside the perpendicular complement is the shuffle arm below, norm-matched per head to $\text{off}_{\perp}$).

The perpendicular component recovers essentially all of the lift (ARC acc_norm 0.8525, ~100%; HellaSwag, Winogrande, TruthfulQA 99–104%), while the parallel component recovers none (ARC acc_norm 0.505 = the Gate-A base exactly, 0%; raw acc 0.5025). Given $\cos(\theta_{\text{mc}}, v_{\text{inf}}) \approx -0.003$, $\text{off}_{\perp} \approx$ the full offset by construction; the informative arms are $\text{off}_{\parallel}$ (an on-axis push at the offset's own scale does nothing) and the shuffle (equal-norm, same subspace, learnedness as the only variable).

Because the *same* harness that yields 100% for the perpendicular push yields only 6.4% for the on-axis W_{know} reference, the on-axis failure is a real property of the geometry, not an insensitivity of the injection method — the harness is direction-sensitive. This closes the non-identifiability objection [34] on the *harness* side.

We close it on the *write* side as well: a random direction sampled inside the *same* perpendicular complement, norm-matched per head to $\text{off}_{\perp}$ (sanity check: $|\cos|$ to $v_{\text{inf}} \approx 5 \times 10^{-17}$, $|\cos|$ to the learned $\text{off}_{\perp}$ $\approx 0.047 \approx 1/\sqrt{256}$), recovers ~0% of the lift across three seeds (ARC fraction −0.04 / −0.01 / −0.01; the lone non-zero, Winogrande ~12%, is a small lift in noise) against $\text{off}_{\perp}$'s ~100%. Same harness, same norm, same subspace — the only variable is *learnedness* — so it is the learned structure *inside* the off-axis complement that lifts, not orthogonality-plus-norm. Following the recommended rebuttal template of comparing a learned direction against a random one [11], both orthogonal controls (the on-axis reference and the shuffled perpendicular) have an effect on the lift indistinguishable from zero while the learned off-axis push does not; and reported specificity failures of activation directions are *cross-model*, whereas within-model — our setting — directions remain specific [33], so that result does not subsume this control.

(These two "off-axis" readings are not in tension: $\text{off}_{\parallel}$ 0% is a decomposition of the trained offset in *weight space*, whereas the cosine 0.088 is the *net effect on activations* — different objects on different metrics, both saying the on-axis component does not carry the lift.)

A norm 4.3× the class gap also invites the question whether a linear on/off-axis decomposition is still meaningful at the offset's operating point. The dose–response sweep (Section 5.7) is the empirical check: the response to scaling the offset over $s \in [0, 1]$ is smooth and monotone — the scoring lift saturates, then the generation damage onsets — with no discontinuity or chaotic regime anywhere on the axis, so the injection operates on a graded response surface where the decomposition is interpretable. We nonetheless scope the geometric claims to the trained offset at its trained scale rather than asserting magnitude-invariance.

A training-time constructive control closes the causal argument from the other side (necessity, complementing this sufficiency): an on-axis training penalty toward v_{inf}, swept over λ, does *not* rescue transfer at matched lift (fork (b), Section 8) — so the +35 is the off-axis mass by necessity, not only by construction. Together, the $\text{off}_{\perp}$/random spectrum (sufficiency) and fork (b) (necessity) make the off-axis reading a two-sided causal claim.

5.6 No single-layer locus: the lift scales with total offset magnitude

The recovery has no causal layer locus. Across a layer-band patching sweep, the recovered fraction correlates with the *total magnitude* of the learned offset ($\Sigma|r|$) at **0.985**, far more than with the number of heads patched (0.832). (We read this as *no single-layer locus* rather than a strict magnitude-over-head-count causal claim: 0.985 and 0.832 are coupled correlations from the same band sweep, not a test that isolates magnitude from head-count; what the sweep licenses is that no privileged layer repairs the readout, not a proof that head-count is causally inert.) The natural temptation — to read "layers 36–40 carry 87% of the lift" — is a head-count artifact: that band simply contains 71% of the adapter's heads (which span L30–55). There is no privileged layer at which the lossy readout is repaired; the recovery is a property of the aggregate magnitude of an off-axis-of-degree push distributed across heads, consistent with the over-scaled offset (Section 5.5) and the coherent-sum logit signature (Section 5.7).

5.7 The deployment pathologies are coupled to injection, not to the lift

Recovering the lift the contrastive way introduces two deployment costs — no transfer to generative CoT (a mechanistic breakage of multi-step reasoning) and retrieval/recall spillover. (A third cost we previously listed — an end-of-turn "collapse" — is retracted at the end of this section as a measurement artifact.) A clean control shows the two surviving costs are coupled to *how* the offset is injected, not to the lift as such. The generic Align-LoRA that recovers most of the lift (Section 5.2) **preserves generation**: gsm8k 0.96→0.95 (n.s.), where the folded contrastive offset destroys it (gsm8k →0.17 baked, and −25 pp adapter-applied; below).

The retrieval spillover, in turn, is consistent with the off-axis push landing on **retrieval-load-bearing heads** — the adapter's heads overlap heads that carry retrieval (in the D.1 SocialIQA adapter: MMLU −19.59 pp, unrecovered by sparsity tuning from K=24 to K=12; correlational, per-head causal test pending). We keep this head-level overlap claim distinct from the Section 5.5 finding that the two functional correctness *faces* are co-located ($\cos v_{\text{inf}} \cdot v_{\text{ret}}$ +0.283): face co-location implies that re-selecting heads within the same band cannot dodge the retrieval substrate, but it does **not** assert that the adapter's off-axis direction aligns with v_{ret} — by Section 5.5 the offset is off-axis to *both* faces, so the spillover is an off-axis perturbation disturbing a shared substrate, not the lift being projected onto the retrieval face. The pathologies are therefore the downstream shadow of folding an off-axis-of-degree perturbation into the weights, not an inevitable price of the lift: a different injection (generic SFT) recovers much of the lift without them.

A *third* injection mode confirms the coupling from the other side: blending the offset in at **decode time** at reduced strength — a milder intervention than folding it into the weights — still buys no generative transfer. Sweeping the blend coefficient ε on gsm8k (n=100, $\varepsilon \in \{0, 0.2, 0.5, 1.0\}$), no interior ε beats the base by more than noise (base 0.97; ε=0.2 → 0.98, ε=0.5 → 0.99, both within 1–2 items of base at n=100 and under the pre-registered 2pp window), while full-strength decode-time injection (ε=1) reproduces the loss (0.87, −10 pp) — there is no transfer window, so the off-axis lift is not deployable to generation even as a decode-time steering vector. (Its ε=1 −10 pp is milder than the −25 pp of the fully-baked adapter — the blend harness perturbs a subset of heads at generation only — so we read it qualitatively, not as a magnitude.) Even at ε=1 the end-of-turn rate stays at 0.97, independently corroborating the retract below: the generative cost is degraded reasoning, not a failure to terminate.

The contrastive adapter's generative loss on gsm8k is itself a mechanistic loss, not a truncated chain of thought. Run generatively under an apples-to-apples protocol (chat template on, 5-shot, n=400), the adapter scores **0.7275** strict-match against the base's **0.9775** (**−25** pp; R6, the canonical kill-rule run, with an earlier provisional R6probe agreeing at −24 pp). That gap is **flat across a generation-budget sweep**: 0.7275 at caps {512, 2048} (R6probe: 0.7375 flat at {512, 1024, 2048}), with the base **byte-identical** across caps. The base's reasoning runs well under the smallest cap, so the cap never bites it. Turning off the chat template only makes the gap *worse* (base 0.730 / v7mc 0.035, −69 pp): both arms degrade, so the template-ON apples-to-apples figure is the canonical one. The −25 is therefore not a template artifact.

The adapter *does* consume the larger budget — 26/400 of its generations grow with the cap — yet **0/400 documents change strict verdict**. A forensic split of the 105 strict failures at cap 2048 separates the two failure modes: **79** are clean-wrong (they emit a #### answer with the wrong number — reasoning resolved but defective, independent of the cap), and the remaining **26** are genuine non-terminating *reasoning* loops (a real task-specific failure, distinct from the retracted factual-recall artifact below) *already* present at cap 512 (e.g. doc 88: a "… not sold by Harald… " loop filling the budget, no ####), i.e. the failure *precedes* the budget rather than being caused by truncation.

The −25 pp is therefore a mechanistic breakage of the reasoning chain — coupled to the contrastive injection, consistent with Section 5.1's no-transfer — not a measurement artifact of a clipped CoT. These 26 loops are a *task-specific* reasoning failure on gsm8k, **not** evidence of a general termination collapse (which the corrected re-measure below refutes on direct recall).

(We retain the gsm8k caveat: it is OOD and does not measure in-domain transfer, so it bounds the *collateral* claim, not the no-transfer claim of Section 5.1.)

The dissociative triangle. Three facts about the contrastive adapter's generative behaviour dissociate, and we state them together because the gsm8k loss could otherwise be read as a blanket generative failure. (a) *Direct factual recall is clean:* on single-hop recall the adapter terminates at 100% and answers tersely and correctly — the earlier 48→4% / 60→4% "collapse" is a **retracted** eos-token measurement artifact (below). (b) *Multi-step reasoning is broken:* on gsm8k the adapter loses **−25** pp, a mechanistic breakage (flat across a generation-budget sweep; 26 non-terminating reasoning loops already present at cap 512), not truncation and not a termination failure. (c) *The damage is coupled to injection, not to the lift:* the generic SFT that recovers most of the lift preserves gsm8k (0.96→0.95), and the reasoning loss is reproduced across **four independent injection modes** — folded/baked (−25 pp), decode-time blend (ε=1, −10 pp), activation-scale steering (damage confined to the 0.75→1.0 band), and confirmed from the other side by the Align control that recovers the lift *without* the loss — so it is a property of *how* the offset enters the weights, not of the lift.

A confound we state rather than hide: the (a)-clean-vs-(b)-broken contrast conflates two things —

multi-step *reasoning* and *distribution distance* (gsm8k is OOD to the tqa+arc training) — so gsm8k bounds the *collateral reasoning cost*, not the no-transfer claim. The no-transfer per se does **not** rest on gsm8k: it is carried by the per-item arbiter (the fixed ARC items sit at the base's CoT ceiling, 124/124, Section 5.1) and by the out-of-task GPQA cold-MC lift ~0 while base GPQA CoT is 0.71 (Section 4.2). gsm8k carries only the *collateral damage to reasoning*, and that damage is the triangulated four-injection-mode result above.

The collateral reasoning loss is dose-separable from the scoring benefit — in steering, not in the baked adapter. The ε-sweep above varied a decode-time *blend* of one face; a cleaner test places both faces on a *single* magnitude axis by scaling the injected offset itself (activation-scale $s \in \{0, 0.25, 0.5, 0.75, 1.0\}$; s=1 reproduces the adapter's scoring lift — ARC acc_norm 0.505→0.855, +35.0 pp ≈ the baked +35, raw acc 0.5025). On this common axis the two faces cross their thresholds at *different* doses: the scoring benefit **saturates early**, reaching its full value already at s=0.75 (ARC 0.855, identical to s=1.0), while the gsm8k reasoning score holds at base through s=0.75 (0.970→0.950, at the edge of the pre-registered 2pp window) and drops only in the 0.75→1.0 band (→ 0.815). The scoring-benefit threshold thus lies *below* the reasoning-harm threshold, so the 0.75→1.0 magnitude is collateral damage with no accompanying scoring gain (the lift is already at ceiling).

This does **not** reopen a transfer window. We are careful with the wording, because two interior points of the ε-sweep above do print *numerically* over base (ε=0.2 → 0.98 and ε=0.5 → 0.99 against base 0.97): at n=100 those are one- and two-item differences, inside the pre-registered 2pp window, so we read them as noise and not as a gain — no dose produces a reasoning improvement we would defend, but the honest statement is "no dose rises above base beyond noise," not "no dose rises above base." What is dose-separable is the *collateral damage*, not a deployable gain. The separation is a property of the **injected, scaled offset (steering); the −25 pp headline is the fully-baked adapter at** s=1.0, so whether a *trainable* adapter can be held in the low-damage regime is an open question, not a result here (the s-swept gsm8k figures use the steering harness, n=200, and are read as curve *shape*, not against the canonical −25 pp).

We name one prior-art foil directly: an angle–norm account of steering [1] predicts on geometric grounds that a concept-discriminative (angular) readout should saturate at lower magnitude than generative coherence (norm-sensitive) breaks — so a reviewer may read our dose separation as geometrically expected. We distinguish it on three measured grounds that account does not cover: the offset is **off-axis** to the concept direction (Section 5.5, not an on-concept angular push), its mechanism is **non-linear** (the direct-logit reading under-reads it, Section 5.8), and it is **not a norm/temperature rescaling** (cosine with the temperature direction 0.023, Section 5.8) — so the separation here is not the generic angular-saturates / radial-breaks story, and we report it as a dose-refinement of the *collateral cost*, not as a deployable window.

An independent result in a different regime corroborates the *shape* of this dissociation from the write side. A budget-normalized **control-window** law for single-neuron steering [24] finds that coherent behavioral control saturates below a **collapse ceiling**, and — its titular point — that *leverage is not reach*: a single neuron can flip a behavior's readout face (e.g. bypass refusal) in fluent, on-task text while genuine downstream reach appears only later and only for a subset of pivots (three of six audited). That is the single-neuron analogue of our dissociation — control of the *scoring* face at a dose below where generative coherence breaks, with the scoring gain never buying generative reach (Section 5.1) — obtained in an unrelated setting (sparse behavior-gating neurons, not a multi-head correctness offset), so we read it as convergent support for the threshold structure, not as the same experiment.

A previously-claimed end-of-turn collapse is retracted as an eos-token artifact. We earlier reported that the contrastive family collapses end-of-turn termination (V7-mc / D.1 factual 48%→4%, social 56%→4%, an analogous V7-mc 60%→4%). That is a measurement artifact: the verbosity harness (`eval_adaptive_verbosity`) scored termination on the base `<eos>` token, whereas Gemma 4 closes a conversational turn with

<end_of_turn> (token 106). With the corrected turn-stop token, base factual termination is **100%** (reported 48%) and the contrastive adapter's factual termination is also **100%** (reported 4%) — there is *no* collapse; the adapter's factual answers are correct, terse, and terminate ("Paris", "Au", "1945"), and the "degenerate loop" read earlier in this *factual-recall* smoke was junk generated *after* the <end_of_turn> the harness had failed to stop on — a measurement artifact, distinct from the genuine gsm8k *reasoning* loops above.

The honest reframe is that the contrastive adapter **does direct factual recall** (terse, correct, terminating — no degradation vs the base, which is also correct but verbose) and **breaks multi-step reasoning** (gsm8k −25 pp) — which fits the per-item arbiter (Section 5.1: the fixed items are knowledge the base reaches by *reasoning*) and the no-transfer result better than a blanket "cannot terminate" claim did.

The analogous D.1 figures were re-measured: the →4% "collapse" is withdrawn, but a real, modest termination degradation remains on open-ended generation (social 96→76%, creative 68→12%) — adapter-dependent and off the V7-mc headline; this paper makes no →4% termination-collapse claim.

5.8 Mechanism candidates ruled out: not temperature, not token-frequency (a content/symbol signpost)

Two mechanistic accounts of the offset are ruled out, and one is left as a signpost; the exploratory workspace and generation-face analyses that extend this section are collected in Section A and carry no claim here.

Not a temperature / null-space mechanism. A natural hypothesis, given the readout framing, is that the offset acts like a confidence-regulation / entropy neuron — writing into the unembedding null space to rescale the logit temperature without re-ordering the argmax [32]. It does not. A direct-logit decomposition projects the residual write onto the *genuine* temperature direction $d^* = \arg\min \|W_U r - \mathbf{1}\|$ (whose own logit image is 0.9988 uniform, so it is well identified) and finds cosine **0.023** — orthogonal to temperature at chance level (random-direction baseline $\sim 1/\sqrt{d_{\text{model}}} \approx 0.014$). Causal activation-patching at the scoring positions where the lift lives (the option-continuation tokens) confirms it: the **relative** temperature fraction is only **0.215** (CI_{hi} 0.224 ≤ the pre-registered 0.35 kill-rule; in absolute fractions of the effect, temperature 0.177 / ranking 0.803), so the offset re-ranks rather than rescales — it moves logit *differences*, not their mean. *Position matters:* at the next-token generation position the offset flattens the distribution more (temperature fraction 0.38), so the negative is anchored at the scoring positions. (Full decomposition — the d^* identification, the anisotropy caveat on the raw uniform-shift fraction, the causal cosine −0.149, and the linear-attribution-under-reads-the-off-axis-alignment pattern [24] — in Section A.)

Not a token-frequency edit; a content/symbol signpost. Direct-logit attribution over the 48 heads finds **no clean lexical mechanism**; the one robust signature is on the *suppression* side — single uppercase letters, punctuation and structural tokens, exactly the surface tokens of multiple-choice labels and turn formatting — the readable face of the *format-preference* channel that aligned-model calibration distinguishes from answer-decision uncertainty [17]. The most promising unifying candidate is the content/symbol-binding channel MCQA mechanism work isolates [23, 39], but the decisive TEXT-vs-LABEL scoring ablation is **consistent-with, not clean**: on ARC the label-arm lift survives a headroom normalization (Δ_{letter} +0.1575 raw / +0.17 norm) yet the permutation controls that would isolate letter-specificity are ceiling-starved, and on TruthfulQA the letter arm is itself at ceiling (uninformative). We report the symbol channel as a signpost, not a named mechanism. The symmetric cosine test for this third candidate — the same construction standard the temperature and frequency negatives received — has now run: a *mean* structural-token contrast direction (the readout-row difference of means between the ~27k structural tokens and the rest; exact by construction, and its logit image separates the classes cleanly) has cosine **−0.0099** with the offset's aggregate write (**−0.0115** for the uppercase-letter-only variant), at the null floor ($1/\sqrt{d} \approx 0.014$, 3×-null gate 0.041) like the other two; a *full-target* format direction is not linearly identifiable in the unembedding basis at all (identification $r = 0.78 <$ the 0.9 self-gate that d_{freq} passed), consistent with the suppression signature living in the *tail* of the offset's logit image rather than in its mean — which is exactly why the channel stays a signpost: the offset is not the mean format direction either (`runs/oq1_functional_axis/format_control.json`). The *other* Stolfo component — a token-frequency edit — is **closed at the attribution level**: over the full wikitext-103 corpus (121.6M tokens) the recovered frequency direction is identified at r=**0.906** (past its 0.9 self-gate) and the offset's cosine with it is **0.0128** (at the null floor, below the 3×-null gate 0.041), so the offset is not a token-frequency edit. One scope note travels with it: unlike the temperature negative, the frequency control had not received a causal-patching cross-check — and the temperature case showed the causal alignment can exceed the linear read severalfold (0.023 → −0.149) — so we ran that cross-check rather than assume it. It comes back **negative for a sign-flip**: projecting the real base→adapter logit delta onto the frequency signature gives cosine **+0.0059**, CI [−0.0027, +0.0142] — same sign as the direct-logit 0.0128, smaller (×0.46), and crossing zero — where the temperature case flipped sign and grew sixfold (0.023 → −0.149, CI [−0.158, −0.140]). The sharpest contrast is in sign *consistency* rather than magnitude: the per-item frequency projection has comparable mean absolute size to the temperature one (0.120 vs 0.177) but inconsistent sign (66% positive, against 99% for temperature), and a component whose sign varies item-to-item cannot be the mechanism of a lift that is systematic across items. One limit travels with the negative: the frequency direction *orthogonalized against the base logits* — the part of log-frequency not already shared with the temperature channel, which has no direct-logit counterpart — does carry a small but non-zero causal component (cosine **+0.0298**, CI [+0.0221, +0.0374], ∼1/5 of the temperature component and ∼1/27 of the ranking one). Frequency is therefore ruled out as the *mechanism*, not as a trace contaminant.

An exploratory Jacobian-lens analysis (single seed) further places θ_{mc} *off* the model's global workspace — an automatic-channel edit, the mechanistic shape a direction that lifts cold scoring but does not transfer to reasoning should have — and an nnsight causal patch on real generation positions finds the same structure-suppression signature (∼91% structural, a ×3 enrichment); we report both, and the per-position recall-tail resolution, in Section A and rely on them for no claim above.

5.9 Two failure modes: ARC carries the readout story (suppression), TruthfulQA is answer-prior (promotion)

The readout account is established on ARC and must not be read across all tasks uniformly. On **ARC**, the lift is question-conditional: a PMI / decision-conditional control retains **78%** of it (naive +0.3275 → PMI +0.2550), and the choices-only lift is only +0.0375 (CI [−0.005, +0.08], not clearly positive) — so it is genuine, question-conditional, and the case the Section 5.1 arbiter and Section 5.5 spectrum characterize. On **TruthfulQA**, by contrast, much of the +37.25 is an **answer-prior / surface-form Goodhart** effect: the choices-only lift is +0.2225 (CI [+0.1675, +0.2725]) and the adapter selects the correct option **0.5725 of the time with no question at all** (base 0.35, chance 0.229); PMI retains 68%. This matches known multiple-choice shortcut behavior in which highest-probability scoring favors options by surface form independent of the premise [20], and choices-only accuracy sits above chance [3].

The two tasks also fail the cold readout by **opposite calibration mechanisms**, visible in the base gold-rank distribution on the items the adapter flips. On **ARC** the gold is **suppressed** — buried at median rank 2, with only 46% of flipped items having it at rank 1 — so the adapter must *promote* it past more fluent distractors (gold-mass added 0.35 > wrong-mass removed 0.21). On **TruthfulQA** the gold is instead a **flat near-miss**, already at median rank 1 (51% of flipped items at rank 1), so the adapter mostly nudges an already-near-top answer over the line. TruthfulQA is therefore not merely "excluded from the inducibility control" — it is a **second failure mode** of the same lossy readout (a near-miss the answer-prior resolves), distinct from ARC's active suppression. We report TruthfulQA held-out, as evidence the scoring lift is *real* and exploitable, not as evidence for the readout mechanism, which rests on ARC. The paper-level rule: the lift is not a single phenomenon — TruthfulQA is answer-prior (promotion); ARC is question-conditional, off-axis-of-degree, and non-transferable (suppression).

We scope the TruthfulQA result precisely. It was previously a pre-registered hedge — a selection lift on the mc1 metric (+37.25, paired-significant under exact McNemar, Holm-corrected p_{corr} 6.26×10^{-38}, b=6/c=155) pending a held-out-plus-mc2 control — and that control has now **run** (full set, n=817), removing the hedge in a specific way. The lift **generalizes held-out**: on the ~20% never matched into training (164 UNSEEN of 817), the v7-mc gain is Δ_{UNSEEN} mc1 **+37.8**, indistinguishable from Δ_{SEEN} +36.9, and it holds on the second metric (Δ_{UNSEEN} mc2 **+26.1**) — clearing the pre-registered gate (Δ_{UNSEEN} mc1 $\geq +15$ pp and Δmc2 $> +10$ pp) comfortably. So the +37.25 is *not* memorization of train-on-eval pairs (the matched corpus covers ~79.9% of the eval, yet the held-out gain is undiminished), and it is real beyond the mc1 selection metric. The decisive contrast is with the generic control: a capacity-matched Align-LoRA, which recovers most of the *ARC* lift, **collapses held-out on TruthfulQA** (mc1 Δ_{SEEN} +38.6 $\rightarrow$ Δ_{UNSEEN} +7.9; mc2 +24.1 $\rightarrow$ +7.6) — the signature of memorizing / format-familiarity — which is precisely why TruthfulQA is excluded from the inducibility control (Section 5.2), and which marks the contrastive lift here as the genuine, generalizing one. What the held-out control does **not** do is convert the gain into truthfulness: the *character* of the lift is still answer-prior (choices-only +0.2225, above), so the held-out result shows that the answer-prior behavior *generalizes* — a robust selection behavior, not eval-set recall — not that the model has become more truthful. We therefore report TruthfulQA as a real, held-out, generalizing answer-prior lift on mc1/mc2 — a second failure mode of the same lossy readout — never as a truthfulness gain and never as evidence for the off-axis mechanism.

5.10 Status of the evidence

The claims rest on a converging set of controls, all on Gemma 4 31B IT: the per-item arbiter and the GPQA out-of-task control (Section 5.1); the cold-MC calibration signature (Section 5.1, confident burial under fluency; present-but-swamped, residual AUC 0.574 base / 0.667 adapter); the Align-LoRA inducibility control (Section 5.2); the surface-ceiling and same-space-probe channel results (Section 5.3); the read-only selector (Section 5.4); the geometry spectrum, the direct $\text{off}_{\perp}/\text{off}_{\parallel}$/shuffle decomposition, and the on-axis causal controls (Section 5.5); the no-locus sweep (Section 5.6); and the injection-coupling control (Section 5.7).

The load-bearing experiments have run. The **inducibility attribution** is closed (a matched Align-LoRA recovers ~62% of the ARC lift, flat across rank — a lower bound). The **off-axis-of-degree geometry is causal and identifiable**: a perpendicular push recovers ~100% of the lift, a parallel push 0%, the strongest on-axis reference (W_{know}) 6.4%, and a norm-matched random perpendicular direction ~0% (three seeds) — closing the non-identifiability objection on both the harness side and the write side (Section 5.5). The **same-space probe** recovers ARC base-wrong gold-vs-distractor at AUC 0.955 (CI [0.934, 0.973], null 0.494) against the 0.502 collapsed-output ceiling, with a surface-form control at 0.442 — a *channel* claim (the output-scalar channel is lossy relative to the activations), not "pure miscalibration" and not "the probe beats the adapter" (different information sets — 12288 activation dims vs collapsed logits, so 0.955 is not a contest with 0.830); its diffuseness (a random-48 head set decodes at 0.949) is a read-test decoupled from the adapter's write-specificity (Section 5.3). The **read-only selector** built on that probe recovers the lift undamaged (selector 0.943; Section 5.4) — a read-only-selector capability prior work establishes [19, 27], used here as the read leg, not claimed as new — and the **read ≠ write** dissociation is four-way: read (selector 0.943), on-axis write fails (W_{know} 6.4%), off-axis write succeeds ($\text{off}_{\perp}$ ~100%, shuffle ~0%), and the probe's read direction is geometrically aligned with v_{inf}, not with θ_{mc} (moderate, Section 5.4). The no-transfer is corroborated **out-of-task** on GPQA (cold-MC base 0.4747 $\rightarrow$ adapter 0.4697 = -0.5 pp, CI crosses 0, while base CoT is 0.7071). On TruthfulQA the held-out control has run: the lift generalizes (Δ_{UNSEEN} mc1 +37.8) where a generic control collapses (+7.9), so it is real and not memorization, but answer-prior by character (Section 5.9). And the temperature/null-space account of the offset is ruled out **causally** (activation-patching: at the scoring positions the relative temperature fraction is 0.215, with absolute fractions temperature 0.177 and ranking 0.803; the offset moves logit *differences*, not the mean — Section 5.8). The *effect* replicates cross-family (Qwen 14B: same-sign cold-MC lift on all four tasks — three clearly, with Winogrande +3.0 small and not individually significant at n=400, counted as marginal — ARC +18 / HellaSwag +7.8 / TruthfulQA-mc1 +15.8 / Winogrande +3, magnitude below Gemma; null==base gate exact), removing "single-model" from the effect; its off-axis geometry replicates too (whitened cos −0.002, frac⊥ 97.5%; below).

A note on multiplicity and sample-sharing scopes these statistics. The paper's tests fall in two families, and only the first carries formal correction. The *confirmatory* family — the four headline lifts and the pre-registered kill-rules (R2, R6) — is Holm-corrected (m=4, Section 4.1). The *mechanism* family (arbiter, calibration, PMI-residual, probe, selector, geometry arms) we read as converging description rather than as a set of independent significance claims, and no conclusion in it rests on an uncorrected marginal p-value: every load-bearing p is either far below any correction over the full campaign (selector 3.7×10^{-47}; negative control 1.4×10^{-4} and its premise-file robustness 3.6×10^{-5} — the weakest load-bearing figure, the all-ARC contrast $p = 0.0015$, still survives a Bonferroni correction across all ~25 tests run) or is *interpreted as a null* (W_{know} $p = 0.136$), where multiplicity cuts the other way. These diagnostics also share a sample: the arbiter, calibration, probe, selector, and geometry arms are all computed on the vinf_causal n=400 draw and its nested subsets, so their agreement is consistency across facets of one sample, not independent replication. What is independently replicated is the headline magnitude (the A1 run; the R4 full-split run, every delta inside the n=400 CI; the Qwen family) — and, as the corresponding strengthening on the mechanism side, **the arbiter has now been re-run on the full ARC-Challenge test split** (n=1172, outside the shared n=400 draw). It replicates: the base scores 0.4932 and the adapter 0.8558 (+**36.3** pp, against +35.0 on the n=400 slice), the adapter fixes 444 items, and the base already answers $\mathbf{442/444} = \mathbf{0.9955}$ of them with chain-of-thought (CI $[0.984, 0.999]$, against 124/124 on the slice, CI $[0.970, 1.000]$). The negative control replicates with far more power: items neither arm fixes are answered at $\mathbf{134/150} = \mathbf{0.8933}$ (CI $[0.834, 0.933]$), against the fixed items' 0.9955 (Fisher OR 26.4, $\mathbf{p = 1.4 \times 10^{-8}}$, against $p = 1.4 \times 10^{-4}$ on the slice) — so the arbiter discriminates, and its near-ceiling reading on the fixed items is not an artifact of the sample it shares with the other diagnostics. The power caveat is unchanged: ARC remains near its chain-of-thought ceiling, so this confirms the benign reading without distinguishing "recovers knowable" from "fabricates" as a harder benchmark would.

Items resolved this pass: the canonical gsm8k loss magnitude is **−25** pp (R6, base 0.9775 vs v7mc 0.7275, apples-to-apples), flat across a generation-budget sweep → a *mechanistic* loss (multi-step reasoning breakage) and not truncation (Section 5.7); the temperature/null-space negative is now **causal** (patching, temperature fraction 0.215 at scoring positions — Section 5.8); and the effect replicates cross-family (Qwen 14B, 4/4-sign lift). Items still pending, marked here rather than asserted: the ~62% generic fraction is a **tight lower bound** — the Align-LoRA best-effort fairness control ran (r512 recovers *less*, not more, so the contrastive-specific residual ~38% is a firm upper bound); the present-but-swamped claim's base-arm residual CI lower bound (0.520) is *intermediate* and rests on the adapter arm (0.584 > 0.55) — the power increase ran and does not rescue it: no defensible pool, including the adapter-independent $S_{\text{cot\_coldwrong}}$ (CI lower bound 0.512) and the pro-gate union (0.535), clears 0.55, so the base channel is frozen intermediate; the earlier end-of-turn "collapse" figures are **retracted** as an eos-token artifact — the corrected re-measure shows no collapse, with only a modest, adapter-dependent open-generation degradation in the D.1 re-measure (full account and corrected numbers in Section 5.7); the logit-lens frequency control (Section 5.8) — flagged pending in earlier passes — is now **closed at the attribution level** (over the full wikitext-103 corpus the frequency direction is identified at r=0.906, past its 0.9 self-gate, and the offset's cosine with it is 0.0128 at the null floor, so the offset is not a token-frequency edit; its causal-patching cross-check has now run and confirms it — no sign-flip, causal cosine +0.0059 with a CI crossing zero, Section 5.8); and the read-only selector's feature set is item-held-out but not feature-held-out, pending the cross-benchmark check (ARC-Easy / OpenBookQA). The cross-family universality of the *effect* now has one replication (Qwen 14B, 4/4 sign), and the off-axis-of-degree *mechanism* now replicates too (Qwen whitened $\cos(\theta_{\mathrm{mc}}, v_{\mathrm{inf}})$ −0.002 within null, perpendicular fraction 97.5%; above). The collateral reasoning loss is now shown **dose-separable from the scoring benefit in steering** (single activation-scale axis: the scoring lift saturates at s=0.75 where gsm8k is still at base, and the reasoning damage falls only in the 0.75→1.0 band — Section 5.7), a dose-refinement of the *collateral* cost (not a deployable window: no dose lifts reasoning above base beyond noise, Section 5.7) whose dissociation-of-thresholds claim survived a prior-art gate (no-scoop; the nearest foil, an angle–norm account [1], is distinguished in Section 5.7), with whether a *trainable* adapter inherits the low-damage regime left as open (steering ≠ baked).

6 Conclusion

We asked what a large cold multiple-choice lift from contrastive-correctness PEFT is, on a frontier 31B instruction model, and the answer is that it is not a capability gain. A capacity- and budget-matched non-contrastive control attributes most of the lift — ~62% of the ARC gain — to ordinary fine-tuning rather than to the contrastive objective. The recovery lives in a learned direction that is off-axis of degree to the model's own correctness geometry, and that geometry is causal and identifiable on both the harness and the write sides. And the lift does not become generation: the items it fixes already sit at the base's chain-of-thought ceiling, a read-only selector on the same activations recovers it without the deployment costs, and — most plainly — the base asked to emit the answer letter in a single forward recovers more of the base-wrong items (0.970) than the adapter (0.763) at the same cost, tying a reasoning baseline that costs a thousand times more. The effect and its off-axis geometry replicate on a second model family — though only partly: the cross-family lift is same-sign on all four tasks but marginal on Winogrande at n=400, and the causal spine that carries the attribution (the arbiter, the inducibility control, the recovery spectrum) remains single-model.

The practical consequence is a caution for the contrastive-PEFT-for-multiple-choice program. A cold multiple-choice gain of this size can be an off-axis, non-transferring edit to a lossy readout — cheaper to obtain with a read-only selector, and beaten by a reasoning baseline — so a benchmark improvement of this kind is not, on its own, evidence of a better model.

7 Limitations

This paper characterizes a single observable: contrastive-correctness PEFT (the LoFiT/ITI family) recovers a large cold-MC lift that is **mostly inducible by generic fine-tuning** (~62% of the ARC lift), along a learned **off-axis-of-degree** direction that does not transfer to generation and, recovered this way, introduces deployment pathologies. The recovery exploits a fact established in prior work and confirmed here — that cold multiple-choice scoring under a chat template is a **lossy** readout (present-but-swamped, not empty) of knowledge the model already has. The scope is deliberately narrow: a mechanistic study of that effect, not a routing architecture. Several limitations bound even this restricted claim.

The headline numbers are pre-registered with kill-rules; the anti-floor and gen-face rules have now executed. The scoring lifts and the gen-face losses are pre-registered with explicit kill-rules and matched-control designs in our evaluation protocol. Throughout this paper, "pre-registered" means a dated protocol document versioned in the project repository *before* the corresponding run (git-timestamped, released with the code): the headline kill-rules and matched-control designs (protocol of 2026-06-22, runs from 2026-06-27), the arbiter negative-control thresholds (2026-07-15, run 2026-07-21), and the on-axis-penalty lift-matched gate (2026-07-08, run 2026-07-09) — internal and verifiable from the release, not a third-party registry. The anti-floor 5-shot kill-rule (R2, n=400) ran 2026-06-27 and the lift survived, but the rule was pre-registered without fixing *which* residual instantiates it, so we state the formula explicitly: the anti-floor quantity is **a@0 − b@5** — the zero-shot adapter against a fully-prompted base — giving ARC +**22.0** pp, HellaSwag +**10.0** pp, Winogrande +**8.25** pp. It is distinct from the matched-shot quantity a@5 − b@5 (ARC +22.25 pp, McNemar b=99/c=10, $p \approx 10^{-19}$; HellaSwag +17.5 pp, Winogrande +13.75 pp), which answers whether the adapter still wins once few-shot is granted to both arms. The +35 is not a zero-shot-floor artifact under either reading, but the margin is strongly task-dependent — few-shot closes 37% of the ARC headline, 41% of HellaSwag's and 57% of Winogrande's — so Winogrande is the weakest anchor under the anti-floor reading (+8.25 pp), and we do not cite its +13.75 as an anti-floor figure (TruthfulQA carries no anti-floor control at all: lm-eval fixes num_fewshot=0). All four MC anchors are paired-significant beyond the anti-floor control — an exact McNemar test on byte-identical base/adapter items gives ARC $p = 1.32 \times 10^{-32}$, TruthfulQA $p = 1.57 \times 10^{-38}$, Winogrande $p = 1.99 \times 10^{-12}$, HellaSwag $p = 1.63 \times 10^{-8}$, and all four survive Holm–Bonferroni correction ($p_{\text{corr}} \le 1.63 \times 10^{-8}$); none is dropped. This certifies the effect exists and is paired-significant, not that it is on-axis — the on-axis recovery ceiling is 6.4% (W_{know}, Section 5.5). The full-set Δ is now confirmed (the v7mc full-set run R4 closed 2026-06-27: ARC acc_norm +36.2, HellaSwag acc_norm +19.2, Winogrande acc +20.1, TruthfulQA acc +37.1, every delta within the n=400 CI — so the n=400 headline values are conservative, not inflated). The gen-face canonical gsm8k kill-rule (R6) has now run — −25 pp, flat across the generation budget, so the loss is mechanistic and not truncation — and the EOS re-measure has run too, with a consequential result: the earlier end-of-turn "collapse" figures are **retracted as an eos-token measurement artifact** (the verbosity harness scored the wrong turn-stop token; corrected, the adapter terminates factual recall at 100%), so we make no termination-collapse claim; the D.1 re-measure shows only a modest, adapter-dependent open-generation degradation — the full account and every corrected number are in Section 5.7. The gsm8k loss magnitude, previously run-dependent (−5/−17/−25/−78 pp), is now canonically **−25** pp (R6). The *direction* of each effect is robust; the remaining un-certified magnitude is the contrastive-specific fraction (below).

Reproducibility and provenance: the lift is not a leakage or input artifact, and the headline mixes two scoring metrics. Four provenance audits each independently *strengthen* the effect rather than threaten it. (i) *Test↔train overlap is negligible.* On ARC, near-duplicate contamination between the eval set and the adapter's training data is 1.0% (4/400 at Jaccard ≥ 0.8, robust across three matching methods); the stricter genuine-leak rate (matching stem + options + answer) is ≤ 0.5% (2/400), and exact eval-ID overlap is zero. This sits at the clean side of our pre-registered gate (< 1% clean / 3–5% retract), far from the retract line — we report it honestly rather than claiming zero. (ii) *The transfer benchmarks are genuinely held out.* The offset was trained on only two datasets — TruthfulQA and ARC (verified across three independent artifacts: the offset config, the head-probe record, and the eval config; top_k = 48, 800 steps, val_acc 0.94 — a training monitor computed, due to dataset ordering in the validation slice, on TruthfulQA items only, so checkpoint selection never saw ARC signal; Section 3.1) — so HellaSwag, Winogrande, gsm8k, and MMLU never entered training (Winogrande has no training builder at all). Their lift is therefore genuine cross-task transfer, not in-distribution recall. (iii) *Base and adapter see byte-identical inputs.* For every task the base and adapter prompts are byte-for-byte identical across all 400 items (1600/1600 per task), with zero truncation (max ≈ 228 tokens ≪ the 4096 context) — the arms differ only in the inference-time hooks, eliminating an "input distribution differs" confound. (iv) *The setup is deterministic and pinned.* The offset resolves to a single artifact across 8/8 runs (sha256 `fcf74ffc...`; the per-shard safetensors hash is pod-only and not reproducible locally), under transformers 5.8.0 and lm-eval 0.4.12. One honest reporting caveat carries through to the headline metric itself: Table 3 mixes two cold-MC metrics — acc_norm for ARC (+35.0) and HellaSwag (+17.0), and plain acc for TruthfulQA-mc1 (+37.25) and Winogrande (+19.0), because mc1 and Winogrande do not define a length-normalized variant. The choice is not cosmetic: on HellaSwag acc_norm (+17.0) is *smaller* than acc (+22.25), so the headline is the more conservative figure and the lift is not a length artifact. We state the per-task metric explicitly rather than imply a single unified score.

The contrastive-specific residual is a firm upper bound; the fairness control has run. Our inducibility result — that a matched non-contrastive SFT recovers ~62% of the ARC lift — used a single-shot Align-LoRA, and a best-effort fairness control has now run: raising the rank to r512 recovers *less* (frac 0.22 at lr1e-4 / a below-base broken adapter at lr2e-4), so it does **not** raise the generic fraction and the contrastive-specific remainder (~38%) is a firm upper bound on what the contrastive objective contributes beyond generic fine-tuning. The ~62% is thus a tight lower bound, not an under-fit floor. We also note the inducibility control is clean only on ARC: TruthfulQA was excluded as train-on-eval leakage (held-out, the matched control collapses 98%→30%), so the generic-fine-tuning claim is an ARC result and we do not generalize it to TruthfulQA.

The inducibility control shares its corpus with the evaluation, so "generic" is not cleanly separated from "domain exposure." The matched Align-LoRA trains on the same TruthfulQA+ARC budget the effect is measured on (Section 3.6). That leaves the ~62% consistent with two readings we cannot separate here: that ordinary fine-tuning of *any* kind recovers most of the lift, or that fine-tuning *on this domain* does. Separating them needs an out-of-domain SFT arm — a matched control trained on unrelated data — which we did not run: it is a full training run to refine an attribution whose interval is already $[45\%, 85\%]$, so it would sharpen the reading rather than overturn it. We flag it as the natural strengthening rather than quietly claim the stronger of the two readings.

The headline anchors are knowledge benchmarks with low chain-of-thought headroom — by design, and stated as scope. All four headline tasks sit at or near the base's CoT ceiling (ARC 0.98, HellaSwag 0.83, Winogrande 0.9425). That is not an oversight but the condition that makes the per-item arbiter informative — there must be a latent, reachable answer for cold scoring to misrank — and it is what the mechanism predicts the exploit needs: on GPQA, the one hard anchor, with genuine reasoning load and ~23 pp of cold-scoring headroom, the same adapter recovers approximately nothing (Section 4.2). What we have not run is the converse arm: an adapter trained *in-distribution* on a hard, non-saturated benchmark (e.g. MMLU-Pro), asking whether contrastive-correctness PEFT produces a large off-axis lift where the base has few buried-but-reachable answers. The mechanism predicts it does not — the exploit amplifies latent misranked knowledge, and where there is none there is nothing to amplify — and the out-of-task GPQA result is consistent with that prediction, but the in-distribution version is future work, not a result here. Claims in this paper are accordingly scoped to the anchors measured.

Pretraining contamination is not audited — and the claims are robust to it by construction. The leakage audit above covers the adapter's fine-tuning corpus, not the base model's pretraining corpus, which is closed to us; ARC, TruthfulQA, HellaSwag, and Winogrande are public benchmarks and plausibly present in any frontier pretraining mix. Two features of the design make the conclusions insensitive to this. First, every comparison is paired: both arms share the base model, hence whatever pretraining exposure exists, and the deltas subtract it out. Second, the mechanism claims are agnostic about the *origin* of the knowledge: the arbiter establishes that the base reaches the fixed items' answers by reasoning while cold scoring misranks them — equally true whether that knowledge came from generalization or from memorized benchmark text — so the attribution of what the adapter's lift *is* does not change. What pretraining contamination could inflate is the base's absolute CoT ceiling, which we use descriptively, not as a claim.

Single-adapter: this is the anatomy of one training run. We flag single-model (Gemma 4 31B IT) and single-seed (the J-lens and dose analyses) elsewhere, but the sharpest restriction is narrower and we state it plainly: every result here characterizes **one** V7-mc adapter, trained once. There are no training replicates of the adapter under study — no seed sweep over the contrastive run itself — so the 100% recovery anchor, the cosine-0.088 net-effect geometry, and the −25 pp gsm8k loss are properties of that single artifact, and we cannot report how far each would move under a re-train. The replication we *do* have varies the comparison arms (three random-direction seeds, three selector seeds, the rank sweep), not the adapter. The same restriction applies to the head set: every geometric and causal number is a property of the $K = 48$ slice this adapter trains — a hyperparameter inherited from the LoFiT recipe (Section 3.6), with no K-ablation — so the claims characterize this adapter's offset, not Gemma's correctness circuitry at large. Read the specific magnitudes as one draw; what the converging controls support is the qualitative structure — mostly inducible, off-axis, non-transferring.

The "signal lives upstream" claim is a *channel* claim, now sharpened by the activation-probe — and it remains a channel claim, not "pure miscalibration." Our surface-ceiling result (Section 5.3) — a supervised, held-out re-scoring of the output scalars caps at AUC 0.574 against the adapter's 0.830 — is information-set-confounded: the adapter mutates activations while the ceiling re-weights collapsed scalars. The in-space activation-probe that closes this confound has now **run** (no fine-tuning, per-head `o_proj`-input, held out by item, same cold-MC option-continuation distribution): it recovers ARC base-wrong gold-vs-distractor at **AUC 0.955** (CI $[0.934, 0.973]$, null 0.494) against the **0.502** collapsed-output ceiling, with a surface-form control at **0.442** (≈ chance). This widens and cleans the channel gap (0.502→0.955) and licenses a **sharpened channel claim** — the cold-MC output-scalar channel is lossy relative to the activation channel — and nothing stronger. It does **not** become "pure miscalibration / signal intact upstream": a linear probe decoding correctness ∼0.95 from activations is near-universal in capable LMs, so it shows the information *exists*, not that the deployed readout merely under-uses it. Nor is it "the probe beats the adapter" or a representational edit — the probe reads 12288 activation dimensions while the adapter is constrained to nudge collapsed output logits, so 0.955 is not on the same axis as 0.830. The diffuseness the probe reveals (a random-48 head set decodes at **0.949** ≈ the adapter's heads) is a **read-test** that the channel is lossy model-wide, decoupled from the **write-tests** that establish the adapter's specificity (a random norm-matched offset recovers ∼0% of the lift, Section 5.5); it does not undercut the adapter story. We do not generalize the channel claim past ARC: TruthfulQA's probe reaches 0.974 but its surface-form control is non-trivial (0.639, with a gold-first position artifact at 1.000), so it is anchored out of the channel claim and read as answer-prior (Section 5.9).

The no-transfer is corroborated on GPQA but the verifier-guided arm is untested by design. We establish no transfer on ARC (the fixed items are at the base's CoT ceiling) and confirm it on GPQA, where the base reasons successfully (CoT 0.70) yet a scoring shortcut still has no generative headroom. Because the base already reasons GPQA out, a verifier-guided deployment (base generates, adapter scores) has nothing to harvest and we did not build it; that is a design decision following from the result, not an independent test, and a setting with genuine selection headroom (which we did not find here) would be required to revisit it.

The off-axis verdict is of degree and method-dependent, established geometrically and causally on Gemma — with the off-axis geometry now replicated on a second family. The recovery is off the functional correctness direction by a *method-dependent* amount: the random control recovers ≈0% (the direction is learned, not arbitrary), the strongest on-axis reference recovers 6.4%, the contrastive net effect sits at cosine 0.088 and generic SFT at 0.199 (2.3× more on-axis). We deliberately do not claim "purely off-axis" or "off-axis universally" — a weight-space proxy suggested that and the clean activation experiment refuted it. The *causal* spectrum (W_{know} recovery, arbiter, inducibility) is single-model (Gemma 4 31B IT), but the off-axis *geometry* itself now replicates on a second family: on Qwen 14B the cold-MC effect is 4/4-sign and its offset's whitened $\cos(\theta_{\text{mc}}, v_{\text{inf}})$ is −0.002 (vs Gemma's −0.003, both within the null band) with 97.5% of the offset energy perpendicular to v_{inf}, so the off-axis-of-degree direction is cross-family, not single-model — with one honest heterogeneity of degree: Qwen's *raw* on-axis energy budget (2.47%, z+8.9 vs null) is ~4× Gemma's (0.59%), though the covariance-controlled whitened cosine is identical (Section 5.10).

The lift is task-heterogeneous, and TruthfulQA carries a different mechanism than ARC. The ARC +35 is question-conditional (PMI retains 78%; choices-only +0.0375, CI crossing zero) and carries the readout story. TruthfulQA's +37.25 is largely answer-prior surface-form Goodhart (choices-only +0.2225; correct option 0.5725 with no question, base 0.35). Any blanket statement that "the +35 is off-axis" or "the +35 is surface-form" is wrong; the mechanism is task-conditional and our readout/off-axis claim is anchored on ARC.

The direct-logit reading has no clean lexical mechanism; the causal patching test has now run, and the symbol-channel test came back mixed. A logit-lens analysis finds no clean lexical story; the one robust signature is suppression of structural single-uppercase tokens, the readable face of the format-preference channel (not of a termination collapse — that claim is retracted, Section 5.7) rather than an explanation of the lift. The frequency control is now closed at the attribution level: over the full wikitext-103 corpus the recovered frequency direction is identified at r=0.906 (past its 0.9 self-gate) and the offset's cosine with it is 0.0128 (at the null floor, below the 3×-null gate 0.041), so the offset is not a token-frequency edit — and the causal-patching cross-check (which the temperature negative did receive) has now run and does not overturn it: no sign-flip, causal cosine +0.0059 with a CI crossing zero, against the temperature's 0.023 → −0.149 (Section 5.8). The activation-patching test of the offset has now run: it confirms the temperature negative causally (temperature fraction 0.215 at scoring positions), but the decisive TEXT-vs-LABEL symbol-channel test is only *consistent-with*, not clean (mixed across ARC and TruthfulQA, Section 5.8), so the unifying mechanism is still a signpost, not a named channel.

8 Open questions and future work

The within-format difficulty control and independent base-CoT for HSwag/Wino have now run — "out-of-task, not difficulty-bound" is supported. The activation-probe in-space — the decisive fairness control for the channel claim — has **run** (Sections 5.3 and 5.10): it lands at AUC 0.955 against the 0.502 collapsed-output ceiling, resolving Section 5.3 to the sharpened channel claim (the output-scalar channel is lossy relative to the activation channel), not to "pure miscalibration" or a representational edit. The two controls that previously left the difficulty-vs-distribution question open have now run as well (Section 5.1). (a) The *within-format* difficulty split — ARC-Easy vs ARC-Challenge under one cold-MC pipeline — moves the base-wrong fix-rate only **−0.10** (Easy 0.861 → Challenge 0.763) as difficulty rises within ARC, whereas going out-of-task to GPQA collapses it **−0.56** (to 0.202): distribution, not within-format difficulty, drives the collapse. (b) Independent base-CoT for HSwag (0.83) and Winogrande (0.9425) confirms both are high — like ARC's 0.97, well above GPQA's 0.70 — and the recovery is recoverability-gated (the adapter fixes CoT-right items more than CoT-wrong ones: HSwag 0.64 vs 0.31, Winogrande 0.76 vs 0.42), so the +17/+19 lifts also rescue CoT-accessible knowledge rather than exploiting a format shortcut. We therefore now report "out-of-task, not difficulty-bound" as **supported on all three anchors** rather than suggestive — with a standing power caveat: the hard cells are small (HSwag n=32, Winogrande n=12), the Easy→Challenge range stays below GPQA's, and the burial gradient is shallow throughout (corr(fix, z-gap) −0.17). Two further caveats on the GPQA arm remain non-findings and we do not state them as contrasts: recoverability-gating could not be detected there (CoT-right 0.186 = CoT-wrong 0.186 is power-starved at n=43, with 31 unparsed items contaminating the hard cell), and the item-level churn (~21 fixed / ~22 broken) is churn within noise (no McNemar), not compensatory damage. What genuinely remains open is the cross-family universality of this difficulty-vs-distribution reading (below) and the verifier-guided arm (untested by design, Limitations).

A training-time on-axis constraint — the constructive follow-up has now run, and it does not rescue transfer. The experiment that killed the process-reward-model reframe killed an *injected, eval-time* on-axis offset; the constructive hypothesis is instead to train the adapter under a penalty forcing the learned offset toward the functional correctness direction v_{inf}, and ask whether a lift constrained to be on-axis *then* transfers to generative CoT. That penalty has now been wired into training and swept (target v_{inf}, $\lambda \in \{0, 5, 15, 50\}$), against a pre-registered **lift-matched** gate — an on-axis adapter counts only if, at equal MC lift, it retains *more* gsm8k than simply scaling the off-axis offset down, otherwise a merely weaker adapter would trivially do less damage. No λ passes. At λ=5 the on-axis adapter recovers only ARC 0.5725 (against the off-axis 0.8575) while its gsm8k sits 9.9pp (3.8σ) below the off-axis dose curve at matched lift, and the TQA lift is ~0 at every λ (the TQA lift is *entirely* off-axis mass): the on-axis penalty buys no transfer, and at the doses that actually constrain the offset on-axis it kills the lift itself (collapsing toward the W_{know} 6.4% floor). This is **fork (b)** — it *strengthens* the off-axis reading, adding a second causal leg (**necessity**, complementing the **sufficiency** of the $\text{off}_{\perp}$ ~100% / random ~0% spectrum of Section 5.5): the +35 *is* the off-axis mass, not an artifact of injection. It is not under-fitting — λ=0 converges to val_acc 0.92 while λ>0 plateaus flat-and-worse (0.72/0.64), and evaluation used the best-val-acc snapshot, so the on-axis arm was shown its most favorable face and still failed. One door remains untried: a penalty-warmup schedule (W_{know} predicts it will not rescue). A follow-up, not a headline result.

A dose-window efficiency candidate — gated, exploratory. The single-magnitude activation-scale axis of Section 5.7 surfaces an unexpected *positive*: at intermediate doses the offset **shortens** generative CoT without breaking the computation. But it is measured on the injected, scaled *steering* offset (steering ≠ baked), on a single seed, and whether a trainable adapter inherits the efficient regime is exactly the open question fork (b) leaves standing — so we report it as a candidate, not a claim, in Section A.3.

A cheap test-time-compute baseline, narrowed (PMI is already settled). The arbiter implies the adapter approximates knowledge the base already reaches by CoT, which invites the question of whether a cheap test-time-compute baseline Pareto-dominates it on the scoring task. One leg is already settled and does **not** dominate: PMI / decision-conditional re-scoring caps at base-wrong AUC 0.574 against the adapter's 0.830 and actively *hurts* the adapter when applied (Section 5.3), and a same-scaffold DC-PMI head-to-head leaves the adapter recovering items PMI does not (TQA, 28 items, CI lower bound +0.622). Two more legs have now **run** and are reported as the **deployment triangle** in Section 5.4. The *reasoning* TTC family — self-consistency over sampled CoT — recovers 0.970 on the 198 base-wrong ARC items, dominating both the read-only selector (0.914) and the adapter's cold-MC scoring (0.763). And the *same-cost* baseline the review asked for — the base emitting the answer letter in a single greedy forward, no CoT — also recovers **0.970** (192/198), at one forward: it ties the reasoning baseline at ~1000× less cost and dominates the selector and the adapter at equal cost. So the adapter loses to a reasoning baseline *and* to the most trivial same-cost alternative, and the selector's value is evidentiary, not deployment (it too is dominated by letter emission).

The one scope caveat is that letter emission is a lettered-MCQ readout distinct from the cold-MC continuation channel (Section 3.2), so this contrasts two readouts of the same base — the lossy-readout thesis in its sharpest form — rather than a like-for-like scoring test. And on the mc_only set — the items the adapter fixes in cold-MC that v_{inf} does not — self-consistency reaches **124/124**: the Section 5.1 arbiter taken to its extreme, the base reaching by reasoning every item the adapter "fixes" by cold scoring.

Cross-family universality of the mechanism. Every causal and geometric result here is on Gemma 4 31B IT. The behavioral *effect* now replicates concretely on a second frontier family: Qwen 2.5 14B IT, run through the identical pipeline (null==base gate exact), shows a same-sign cold-MC lift on all four tasks (ARC +18.0 / HellaSwag +7.8 / TruthfulQA-mc1 +15.8 / Winogrande +3.0 — the last small and not individually significant at n=400, counted as marginal; magnitude below Gemma's 35/17/37/19), removing "single-model" from the *effect* (three tasks clearly, Winogrande marginally) — and the per-task pattern differs (Winogrande +3 vs Gemma's +19), a difference worth explaining. What remains single-model is the *mechanistic* characterization — the arbiter, the inducibility control, the geometry spectrum, the no-locus result — and, most directly, **whether the Qwen lift is off-axis-of-degree in the same way — which has now run**: extracting Qwen's v_{inf} and Fisher-LDA W_{know} (ARC+HellaSwag inference family, chat-template, 48×40×128 grid) and decomposing its offset gives whitened $\cos(\theta_{\mathrm{mc}}, v_{\mathrm{inf}}) = -0.002$ (within null, ≈ Gemma's −0.003) and a 97.5% perpendicular energy fraction — the off-axis-of-degree geometry **replicates on a second family** (Qwen), lifting it from single-model to cross-family-replicated — suggestive of, but not established as, a general property (N=2). The one cross-family difference is a minor heterogeneity of *degree* (Qwen's raw on-axis budget 2.47% vs Gemma's 0.59%, though the whitened cosine matches), not of sign. What stays single-model is the causal recovery spectrum (W_{know} 6.4%, arbiter, inducibility).

Open questions inherited from the system-level negatives. Two further questions bound the deployment reading rather than the mechanism. The false-positive routing gate is unbuilt (the only damage-screen is weak, AUC 0.609), so we make no hazard-free deployment claim; and the per-item fine-dispatch nulls were estimated against a permutation floor at n=120 and an underpowered cross-task screen at n=62, so they are reported as no-signal-at-this-power, not established absence. Neither affects the central result, but both must be settled before the routed system is read as more than a mitigation vehicle.

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Code and data availability

Code to reproduce every analysis, the evaluation protocols and their pre-registered kill-rules, and the result artifacts behind each table are released with this paper; the adapter offsets and the heavy activation/evaluation artifacts are hosted as Hugging Face datasets referenced per-number in Section 3.6 (run → artifact map).

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Table 1: Run → population map. Every headline number traces to one of these runs; base-wrong subsets are labeled by scaffold so no two counts are conflated.

Run (artifact)	Date	Scoring n	Base-wrong subset(s)	Feeds
A1 canonical trio (`msap-a1-trio-b2-planner-2026-06-10`)	2026-06-10	400	ARC 198 base-wrong@0-shot	T1 headline (+35.0 / +37.25 / +17.0 / +19.0); T2 gsm8k −25
vinf_causal (`msap-vinf-causal-20260623`)	2026-06-23	400 (own within-run baselines; ARC lift +35.0)	*lm-eval scaffold:* 198 base-wrong; 124 mc_only; 151 S_{flip}; 192 $S_{\text{cot\_coldwrong}}$; 47 adapter-residual. *premise-file scaffold:* 204 base-wrong; 73 adapter-residual	T3 arbiter (124/124, parser-fixed) + its negative control (residual 41/47, `arbiter_negative_control_b2.json`); T3b calibration / present-but-swamped; T6 v_{inf} 11%; T8 PMI-DC / choices-only
wknow_causal (`msap-wknow-onaxis-20260623`)	2026-06-23	400 (null-adapter baseline 0.500)	—	T5/T6 W_{know} 6.4%; significance (McNemar $p = 0.136$)
R2 anti-floor 5-shot (pre-registered eval)	2026-06-27	400	ARC 198 base-wrong@0 (adapter fixes 151; few-shot 69)	T1 caption: anti-floor +22.0 pp (a@0−b@5) and matched-shot +22.25 pp (a@5−b@5; McNemar b=99/c=10)
R4 full-set (`msap-eval-bulletproof-lean-20260627`)	2026-06-27	full test sets	—	full-Δ confirmation (ARC +36.2 / HSwag +19.2 / Wino +20.1 / TQA +37.1)
Same-cost letter baseline (`B5_base_letter_greedy_arc400.json`, same repo)	2026-07-21	400 (198 base-wrong)	ARC 198 base-wrong	base greedy, no-CoT, 1-forward letter emission → 0.980 all / 0.970 base-wrong; deployment triangle 4th point (Section 5.4)
mechanism_v8 (`msap-mechanism-v8-20260701`)	2026-07-01/02	400	—	T2 R6 gsm8k −25 (cap sweep); Section 5.7 EOS retract + Stage-E blend; Section 5.8 patching; Section 5.10 Qwen 4/4-sign

Table 2: Number/analysis → local result file → Hugging Face dataset backup.

Number / analysis	Local file (`runs/...`)	HF dataset (`sebacascardo87/...`)
Arbiter 124/124 (post-parser-fix v2)	`vinf_causal/arbiter_arc_cot_v2.json`	`msap-vinf-causal-20260623`
P9 on-axis-penalty lift-matched gate (λ=5 −9.9 pp / 3.8σ)	`p9_onaxis/{mc,gsm8k}_lam{5,15,50}.json`, `mc_dose_lam0_s*.json`, `gsm8k_dose_lam0.json`	`msap-p9-onaxis-20260709`
Recall-tail (Section 5.8 / Section A.2)	`jspace/stage_e2_recall_tail.json`	`msap-jspace-gemma-20260709`
Align-fairness rank sweep (~62%)	`align_fairness/R1_{v7mc,null}.json`, `phase1/derisk/align_direction_r*.json`	`msap-align-fairness-20260702`, `msap-phase1-20260624`
Table 8 random norm-matched	`phase1/derisk/random_s{0,1,2}.json`, `offsets_random_s*.pt`	`msap-phase1-20260624`
Section 5.5 off$_{\parallel}$ / off$_{\perp}$ / W_{know} spectrum arms	`eval_bulletproof/RD_{offpar,offperp,wknow}.json` (+ `rd_offperp_frac.csv`)	`msap-r5-align-p0-20260628`
Section 5.5 perpendicular-shuffle write-side control (3 seeds)	`RD_offperp_shuffle_seed{0,1,2}.json`	same

Table 3: Headline scoring-face MC lift (V7-mc, Gemma 4 31B IT, n=400, chat-template, lm-eval canonical). The cold-scoring lift the whole paper is about. Same flags both arms; never cited without the gen-face row (Table 4) and the protocol caveat (base ≈0.50 is loglik-MC scoring-face, *not* a third-party-reported ~91% gen-CoT figure — a blog, not Google's official technical report). TruthfulQA-mc1 is reported separately as an answer-prior selection lift (mechanism-excluded; it generalizes held-out — Table 11), never as a truthfulness gain and never as evidence for the off-axis mechanism.

Benchmark	Base	V7-mc	Lift	Source artifact
ARC-Challenge (acc_norm)	0.505	0.855	**+35.0**	`msap-a1-trio-b2-planner-2026-06-10`
TruthfulQA-mc1	0.4375	0.810	**+37.25**	same
HellaSwag (acc_norm)	0.540	0.710	**+17.0**	same
Winogrande	0.6575	0.8475	**+19.0**	same

Pre-registered with kill-rules; the anti-floor kill-rule **R2 has executed (**n=400**) and the lift survived**. The paired 5-shot control, its two named quantities (**anti-floor** a@0 − b@5: ARC **+22.0** pp / HSwag **+10.0** pp / Wino **+8.25** pp; **matched-shot** a@5 − b@5: ARC **+22.25** pp, McNemar b=99/c=10, $p{\approx}10^{-19}$ / HSwag **+17.5** pp / Wino **+13.75** pp; few-shot closes 37/41/57% of the respective headlines), the per-item coverage, and the task-specific saturation are reported in Section 4.1. **TQA carries no anti-floor control** (lm-eval `truthfulqa_mc1` fixes num_fewshot=0; both arms 0-shot) → cite the +37.25 separately, with no "irreducible-to-few-shot" claim. The artifact-backed replication is verified.

Paired significance (n=400**).** All four lifts are **paired-significant** by an exact McNemar test computed over byte-identical base/adapter items (prompts byte-identical 400/400 per task, 0 truncation): ARC b=11/c=151 $p = 1.32\times10^{-32}$, TQA b=6/c=155 $p = 1.57\times10^{-38}$, Winogrande b=23/c=99 $p = 1.99\times10^{-12}$, HellaSwag b=39/c=107 $p = 1.63\times10^{-8}$. They survive **Holm–Bonferroni** correction across the four anchors (m=4): p_{corr} ARC 3.95×10^{-32}, TQA 6.26×10^{-38}, Wino 3.98×10^{-12}, HSwag 1.63×10^{-8} — every anchor at or below 1.63×10^{-8}, **none dropped**. *Scope caveat:* this certifies the effect **exists and is paired-significant**, not that it is on-axis — the on-axis ceiling is W_{know} 6.4% (Section 5.5). *Metric note:* the headline mixes acc_norm (ARC/HSwag) with acc (TQA/Wino, which define no acc_norm); on HSwag acc_norm ≤ acc (+17.0 vs +22.25), so the length-normalized headline is the conservative figure, not a length artifact.

Table 4: Gen-face: V7-mc loses on generation (Gemma 4 31B IT). The other half of the trade. The verified anchor is **gsm8k**, the direct "ARC-in-CoT" test (`generate_until`, exact_match on the generation): V7-mc loses in every condition. **Direction (loss) is robust; the canonical magnitude is now** −25 pp (R6 kill-rule PASS: base 0.9775 vs v7mc 0.7275, apples-to-apples template-ON 5-shot, flat across cap {512, 2048} → mechanistic multi-step-reasoning breakage, not truncation). The retrieval-side **lambada** loss (−32) is reported here only as an *illustrative* signal: it comes from a custom token-norm scorer, and the lm-eval confirmation (R9, `lambada_openai` template-off) is not yet run, so the inference/retrieval trade-off in this paper rests on gsm8k alone, not on lambada.

Benchmark	Base	V7-mc	Δ	Source artifact
gsm8k (A1 run, strict-match)	0.9775	0.7275	**−25.0**	A1 run, lm-eval canonical; confirmed R6
gsm8k (day3, mu_only)	~0.955	0.905	−5	`runs/phase4_routed/day3/`
gsm8k (day3, romulo)	~0.955	0.785	−17	same
gsm8k (v7.5-baked, custom eval, *n*=200)	0.955	0.17	**−78**	`runs/v7_5_lofit/eval_gsm8k_v75.json`
gsm8k canonical magnitude	0.9775	0.7275	**−25.0**	R6 (*n*=400 paired, apples-to-apples template-ON 5-shot, cap sweep {512, 2048}=0.7275; template-OFF worse: base 0.730/v7mc 0.035, −69 pp; R6probe agreed −24 pp)
lambada (illustrative)	0.64	0.32	−32 (ppl 11.6→2523)	`runs/v7_lofit_gemma4_31b_chat/eval/` — **custom token-norm scorer**, audit-verified 2026-06-10

Table 5: The fixed items are CoT-accessible; no transfer even on a hard task (the per-item arbiter). The core result. The adapter "fixes" items the base already answers by reasoning; the gain is re-ranking, not capability — and this holds where the base must genuinely reason (GPQA).

Quantity	Value	Source artifact
ARC items adapter fixes in cold MC but on-axis offset does not	124 (mc_only; mc∩vinf=27 / mc_only=124)	`surface_stratified_arc_challenge.json`
Of those 124, base solves under generative CoT	**124/124** = **1.000** (fixed-parser re-run: n_unparsed 5→0; the earlier "123/124" was a parser artifact — doc 128's numeric option label "4" the letter-parser could not emit, now parses correct; ≈ base-right control 0.990)	`arbiter_arc_cot_v2.json`
Negative control — adapter's residual (base-wrong items it does *not* fix)	**41/47** = **0.872** CI[0.748, 0.940] vs **151/151** on the ones it fixes (Fisher $\mathbf{p = 1.4 \times 10^{-4}}$) and 392/400 = 0.980 over all of ARC ($p = 0.0015$) → **the arbiter discriminates**; robustness in the premise-file scaffold 66/73 = 0.904 vs 326/327 ($p = 3.6 \times 10^{-5}$). *Same run, same protocol* (re-partition, no new generation). Point estimate fell between the pre-registered ≤0.85 / ≈0.97 thresholds; reading rests on the contrast, and the absolute magnitude is modest (ARC near its CoT ceiling throughout)	`arbiter_negative_control_b2.json` (`analyze_b2_arbiter_negative_control.py`)
Base-right control (already-correct-cold items)	**200/202** = **0.990** (fixed-parser run; the superseded pre-fix run gave 196/202 = 0.970)	`arbiter_arc_cot_v2.json`
Base GPQA accuracy under CoT (no-transfer is out-of-task, not difficulty-bound)	**0.70** (139/198; thinking-ON greedy, max_new=8192; thinking_rate 1.000) — a separate earlier run; the per-item arbiter run scores 140/198 = 0.7071, differing by one item and supporting the same conclusion	`msap-phase1-20260624 b1_gpqa_base_cot_8k.json`
GPQA cold-MC scoring lift (adapter, out-of-task)	**−0.5** pp (base 0.4747 → adapter 0.4697; CI crosses 0); arbiter base GPQA CoT 0.7071; fix-rate 0.202 over 104 base-wrong	`msap-phase1-20260624`
Reading	fixed items CoT-accessible; cold-MC misranks; no scoring headroom out-of-task (GPQA lift ~0)	—

GPQA note. Base CoT 0.70 > chance → the base *does* reason these out, yet a scoring shortcut still cannot transfer → the no-transfer is not an artifact of ARC being near its CoT ceiling. We therefore did not build a verifier-guided arm (a design choice following from the no-headroom result, not an independent test). With 2048 generation tokens the base truncated 96% of generations → a spurious 0.15; max_new=8192 is required. The GPQA figures are single-run (greedy) and directional (CI crosses 0), not n=400-replicated like ARC.

Table 6: How the readout misranks, and that the lift is not interpretable re-scoring. The *how* of the readout failure (confident burial under fluency) and the channel result (the lift does not reduce to re-scoring the output scalars). All ARC-Challenge, Gemma 4 31B IT, base-wrong items unless noted.

Quantity	Value	Source artifact
Base buries gold confidently — chosen-vs-gold z-gap (adapter-independent $S_{\text{cot\_coldwrong}}$, n=192)	**1.45**σ; gold-rank median **2**; frac rank≥2 **0.59**	`cold_mc_calibration_arc_challenge.json`
Consistency check (S_{flip}, n=151, adapter-conditioned)	z-gap **1.37**σ; gold-rank median **2**; frac rank≥2 0.54	same
Gold-vs-distractor naive AUC (base-wrong, *below* chance)	**0.394** CI[0.356, 0.430], n=204	`residual_discrimination_arc_challenge_base.json`
Base raw pick = gold ($S_{\text{cot\_coldwrong}}$ vs base-right control)	**0.19** vs **0.81**	cold_mc_calibration
Entropy "flatness" is a softmax-temperature artifact	acc_norm answer-entropy 0.920 ($S_{\text{cot\_coldwrong}}$; length-norm compresses spread ~50×); report rank/z-gap, not entropy	cold_mc_calibration
Adapter re-ranking is **promotion-dominated** (S_{flip})	gold-mass +0.35 > wrong-mass removed +0.21; Δentropy −0.19	cold_mc_calibration `re_routing`
Present-but-swamped: PMI-residual AUC (same scalar channel, different populations)	base-wrong **0.574** CI[0.520, 0.627] n=204 *(intermediate, CI_{lo}<0.55)* / adapter residual **0.667** CI[0.584, 0.744] n=73 *(claim rests here)*	`residual_discrimination_arc_challenge_{base,mc}.json`
Surface ceiling (supervised, held-out, base-wrong AUC) vs adapter	logistic **0.458** / GBM **0.462** / on-test oracle stack **0.502** (19% recovery) / PMI **0.574** vs adapter **0.830**	`offaxis_vs_stack_` + `supervised_surface_ceiling_arc_challenge.json`
PMI applied to the adapter / adapter re-rank explained by surface	PMI **hurts** 0.818→0.758; R^2(re-rank|surface) **0.024**	`offaxis_vs_stack_arc_challenge.json`

Reading. The cold readout buries the gold under fluent surface forms (z-gap 1.45σ, naive AUC 0.394 *below* chance), the correctness signal is *present but swamped* (PMI-residual clears chance on the adapter arm 0.667 n=73; base 0.574 n=204 intermediate — **lossy, not empty**), and no interpretable re-scoring of the output scalars recovers the lift (ceiling 0.574 ≪ adapter 0.830; PMI *hurts*; R^2 0.024). The 0.830-vs-ceiling gap is **information-set-confounded** (activations vs collapsed scalars) → a **channel** claim, not a "magic direction" (generic SFT recovers ~62% the same way) nor "empty channel". **The in-space activation-probe has now run and resolves this gate:** a logistic probe on the *same* per-head `o_proj`-input activations (held out by item, same cold-MC option-continuation distribution, preprocessing fit inside each fold, within-item permutation null) recovers gold-vs-distractor on ARC base-wrong at **AUC 0.955** CI[0.934, 0.973] (null 0.494), against the **0.502** collapsed-output surface-cap; a surface-form control on the option string caps at **0.442** (≈ chance) → the 0.955 reads correctness the output scalars discard, **not** a length/format artifact. This **widens and cleans the channel gap** (0.502→0.955) and supersedes the preliminary wrong-space ~0.65. **Three guards hold the read down to a channel claim** (match Section 5.3): (1) the probe reads 12288 activation dims while the adapter only nudges collapsed output logits — different information sets, so 0.955 is **not** "probe beats adapter" and **not** a representational edit; (2) a linear probe decoding correctness ~0.95 is near-universal in capable LMs → it shows the info *exists* in the activations, **not** "pure miscalibration / signal intact upstream" — we keep **lossy** and drop "pure miscalibration"; (3) the signal is **diffuse** — a random-48 head set decodes it at 0.949 CI[0.924, 0.971] (null 0.492) ≈ the adapter's 48 heads → a **read-test** that the cold-MC channel is lossy *model-wide*, **decoupled** from the **write-tests** that establish the adapter's specificity (W_{know} 6.4%, random norm-matched *offset* ~0% recovery of the lift — a different operation; read ≠ write). TruthfulQA is **anchored out** of the channel claim (probe 0.974 but surface-form control 0.639 + gold-first artifact 1.000 ⇒ answer-prior, Section 5.9); the channel claim is anchored on **ARC**.

Table 7: The recovery is mostly generic to fine-tuning (Align-LoRA matched control). A non-contrastive SFT control of matched capacity/budget recovers most of the lift → the effect is **not special to the contrastive objective**. ARC is the clean test; TQA is excluded as train-on-eval leakage.

Quantity	Value	Source artifact
ARC lift recovered by matched Align-LoRA (rank sweep r8/32/64/256)	fraction **59**/**63**/**60**/**65** = ~**62**%, flat	`msap-phase1-20260624` (rank sweep)
Capacity-resolvability of the residual	slope-CI [−**1**.**4**, +**3**.**4**] (includes 0); delta-method frac-CI ~[**45**%, **85**%] (zone C)	same
Best-effort capacity flank (r512, causal fairness, n=400)	lr1e-4 frac **0**.**22** (converged, acc>base; overfit ep2→ep4, robust ep2=ep4); lr2e-4 **below base** (broken, excluded like r256hi) → **does not raise** ~62%	`msap-align-fairness-20260702`
Contrastive-specific residual	~**38**% of the ARC lift (= 100% − ~62%)	derived
TQA excluded — leakage (proof)	frac 41/66/72/91 monotone↑; held-out 20%: r256 **98**%→**30**%; r8=r256=0.524; contrastive generalizes held-out (0.83)	same

Note. The ~62% is a **tight lower bound**. The Align-fairness best-effort control (r512 / LR-sweep / +epochs) **ran** and does **not** raise the generic fraction — more capacity recovers *less* (frac 0.22 at lr1e-4, converged; below-base broken at lr2e-4) → the ~62% is the ceiling of generic-FT capacity and the ~38% residual a firm upper bound. This de-risk is now **closed**.

Interval validation and the r512 flank. Because delta-method (Wald) intervals on ratio estimators can be unstable when the denominator's sampling error matters, the fraction interval is cross-checked by an independent bootstrap-of-the-ratio (200k parametric draws from the published arm accuracies at n=400, independent-binomial resampling — conservative relative to the paired design, whose positive inter-arm correlation would only tighten it): r256 frac CI [0.50, 0.80] against the delta-method [0.45, 0.85], and a bootstrap slope CI of [−2.5, +4.6] pp per log2-rank step containing zero, both matching the published reading (`runs/oq1_functional_axis/frac_ci_bootstrap.json`). The r512 flank is deliberately *not* folded into the flat-sweep fit: at lr 1e-4 it converges yet recovers markedly less (0.22, stable across epochs 2–4, overfitting beyond), and at lr 2e-4 it breaks below base — both excluded by the same convergence-pathology rule that excluded the r256 high-LR arm, so the flank *bounds* the sweep (more capacity does not raise the fraction) rather than extending the regression.

Table 8: Geometry spectrum: off-axis of degree, method-dependent. Apples-to-apples: the *net effect on activations* of each method, decomposed against the functional correctness direction v_{inf}. `ratio` = on-axis alignment of the net effect with v_{inf}, normalized to a random direction (1.0 = like random; higher = more on-axis than chance).

Direction	Recovery of the lift	cos with v_{inf} (ratio vs chance)	Source artifact
random norm-matched (3 seeds)	≈**0**% (acc_norm 0.50–0.52 ≈ base 0.505)	0.05 (1.0)	`msap-phase1-20260624 derisk/offsets_random_s*`
on-axis maximum (W_{know}, Fisher-LDA)	**6.4**% (ARC null 0.500 / mc 0.853 / wknow 0.522; **statistically indistinguishable from zero**: exact McNemar b=10/c=19 $p = 0.136$, paired-bootstrap frac-CI95 $[-0.009, +0.136]$ and Wilson acc-CI $[0.474, 0.571]$ both include chance)	— (cos with v_{inf} 0.379; cos with θ_{mc} 0.0014)	`runs/wknow_causal/` + `msap-wknow-onaxis-20260623`
contrastive adapter (net effect)	100% (by construction)	**0.088 (1.74)**; direct θ_{mc} offset cos 0.0014	`msap-phase1-20260624 derisk/contrastive_direction`
generic SFT (net effect)	∼**62**%	**0.199 (3.93)**	`msap-phase1-20260624 derisk/align_direction`

Verdict. Both methods are **mostly off-axis** (cos<0.2, ∼78–85° from v_{inf}), but generic SFT is **2.3× more on-axis** than the contrastive adapter → **off-axis of degree, not universal; not contrast-special.** The direct θ_{mc} offset (cos 0.0014, a weight-space proxy) was misleading; the clean net-activation-effect experiment is the canonical measurement. **Forward filter:** a zero-cost read-space preview said "off-axis" (ratio ∼1.05); the clean experiment refuted it (3.93). Report only the clean experiment.

Table 9: On-axis causal controls (no on-axis reference reproduces the lift). The earlier causal arms, consistent with Table 8's spectrum: pushing *along* correctness does not recover the lift. (These on-axis causal arms were measured on the vinf_causal run, whose within-run MC baselines — ARC +35.0, TQA +36.5 — differ slightly from the A1 canonical of Table 3, +37.25; fractions are computed within-run.)

Quantity	Value	Source artifact
ARC lift recovered by on-axis v_{inf} (mass-mean, norm-matched)	fraction **0.11** (MC +35.0 / VINF +4.0; $z = -6.63$; acc_vinf 0.545, Wilson CI $[0.496, 0.593]$ includes chance 0.500)	`runs/vinf_causal/` + `msap-vinf-causal-20260623`
ARC lift recovered by W_{know} (strongest on-axis ref)	fraction **0.064** (null 0.500 / mc 0.853 / wknow 0.522; ≪ 0.30 kill-rule; **indistinguishable from zero**: exact McNemar b=10/c=19 $p = 0.136$, paired-bootstrap frac-CI95 $[-0.009, +0.136]$ includes 0)	`runs/wknow_causal/`
TQA lift recovered by either on-axis ref	∼**0** (VINF −2.75; W_{know} ≈ null)	—
θ_{mc} vs v_{inf}	$\cos_{\text{w}}$ **−0.003** (within null); 6.4% mass in top PCs; norm 4.3× class gap (the top-PC 6.4% is coincidentally equal to the W_{know} recovery 6.4%, not the same quantity)	`theta_mc_characterization.json`
MMLU damage, on-axis v_{inf} vs mc	VINF −5.83 ≥ MC −4.30 (gate PASS, null MMLU +0.07 pp)	—

`process-LoFiT-PRM` (an on-axis training program) is **refuted causally** at eval time (a PRM corpus is not warranted); a *training-time* on-axis constraint **has now run and does not rescue transfer** (no λ passes the lift-matched gate, so the +35 is the off-axis mass by *necessity*; see Section 8).

Table 10: No layer locus: the lift scales with offset magnitude, not layer.

Quantity	Value	Source artifact
corr(recovered lift, total offset magnitude $\Sigma\|r\|$)	**0.985**	`msap-phase1-20260624 patch_layers/*.json`
corr(recovered lift, head count)	0.832	same
"Layers 36–40 carry 87%"	head-count artifact (that band holds 71% of adapter heads); adapter spans L30–55 — **not a causal locus**	same

Table 11: Task heterogeneity: ARC ≠ TQA. ARC is question-conditional and carries the readout story; TQA is largely answer-prior / surface-form Goodhart. The held-out + mc2 control has run: TQA generalizes held-out → reported as a real, generalizing answer-prior lift, not as the off-axis mechanism.

Task	Diagnostic	Value	Source artifact
ARC	PMI-DC retention of lift	**78**% (naive +0.3275 → PMI +0.2550)	`pmi_dc_arc_challenge.json`
ARC	choices-only lift	**+0.0375** CI[−0.005, +0.08] (not clear) — question-conditional, not answer-prior	`answer_only_arc_challenge.json`
TQA	choices-only lift	**+0.2225** CI[+0.1675, +0.2725] — answer-prior	`answer_only_truthfulqa_mc1.json`
TQA	adapter picks correct option with NO question	**0.5725** (base 0.35, chance 0.229) — surface-form Goodhart [3, 20]	same
TQA	PMI-DC retention	68%	`pmi_dc_truthfulqa_mc1.json`
TQA	paired significance of the +37.25	exact McNemar b=6/c=155, p_{corr} $\mathbf{6.26 \times 10^{-38}}$ (Holm m=4)	paired doc_hash 400/400
TQA	held-out generalization ≠ memorization (full set n=817; train covers ~79.9%, 653 SEEN / 164 UNSEEN)	v7mc $\mathbf{\Delta_{\mathrm{UNSEEN}}}$ **mc1 +37.8** ≈ Δ_{SEEN} +36.9; mc2 Δ_{UNSEEN} +26.1 (gate Δ_{UNSEEN} mc1 ≥ + 15 AND Δmc2 > + 10 passed); generic Align **collapses held-out** (mc1 +38.6→+**7.9**; mc2 +24.1→ + 7.6)	`analyze_tqa_contamination_l4.py`, full 817

Note. The lift is task-heterogeneous. TruthfulQA is an answer-prior effect — a generalizing selection lift on mc1/mc2 (the held-out + mc2 control generalizes: Δ_{UNSEEN} mc1 +37.8, so it is neither memorization nor mc1-only, while a generic Align control collapses held-out to +7.9) — but it is not a truthfulness gain and not evidence for the off-axis mechanism. ARC is question-conditional, off-axis-of-degree, and non-transferable.

Table 12: Deployment pathologies (coupled to injection method, not to the lift). The recovery introduces **two certified costs** — no transfer to generative CoT (multi-step reasoning breakage) and retrieval/recall spillover — that a generic SFT recovering most of the lift does not incur, so they trace to how the contrastive offset is folded into weights, not to the lift itself. A third, previously-listed cost — an end-of-turn "collapse" — is **retracted** as a measurement artifact (last row). Rows 2–4 are **family** evidence from sibling adapters (the D.1 SocialIQA and B.2/KR adapters), not the V7-mc adapter under study; V7-mc's own directly-measured MMLU spillover is −**4.2** pp (row 5).

Pathology	Value	Source artifact
No-CoT-transfer (multi-step reasoning)	gsm8k loss every condition (Table 4); contrastive baked →0.17. **R6 canonical (kill-rule PASS, apples-to-apples, template ON, 5-shot,** *n*=400**):** base **0.9775** vs v7mc **0.7275** = −**25** pp, flat across cap {512, 2048}=0.7275/0.7275 → **not truncation** (extra budget never rescues; v7mc cap2048 runs ~2h to the ceiling without improving; template-OFF is worse — base 0.730 / v7mc 0.035, −69 pp — so it is not a template artifact). Forensic (R6probe @cap2048): 79 clean-wrong (#### wrong number — reasoning resolved but defective) + 26 degenerate non-terminating loops already present @cap512 (the loop precedes the budget, not truncation). **Align SFT preserves** 0.96→0.95 **(NS)**	R6 (`runs/mechanism_v8/R6_*`) + R6probe (`R6probe_*`) + Table 4
Retrieval spillover (D.1 SocialIQA adapter — family evidence)	MMLU −**19.59** pp (0.8306→0.6347); K=12 not recovered (−20.80, Pareto-worse)	D.1 SocialIQA spillover audit
Spillover (KR SciQ K=24)	GPQA Diamond −**12.1** pp (sciq +2.1 pp overshoot)	KR SciQ K=24 overshoot audit
Spillover (B.2 mc_discrimination)	MMLU −6.3 pp / GPQA −6.6 pp	B.2 phase-3 findings
Spillover MMLU (V7-mc direct, eval-bulletproof R10)	−**4.2** pp (0.8226→0.7805, ~2k stratified, 0-shot) — the V7-mc adapter under study spillovers MMLU itself; same sign as B.2 (−6.3 pp), smaller	`runs/eval_bulletproof/R10_*`
Spillover mechanism	$\cos_{w}(v_{inf}, v_{ret})$ +**0.283** BCa95[+0.276, +0.291], null ±0.007, *n*=316 — orthogonality refuted; co-location on a shared substrate	functional-axis geometry analysis
~~End-of-turn collapse~~ **retracted — eos-token artifact** (was: D.1 social 56→4, D.1 factual 48→4, V7-mc factual 48→4/60→4)	`eval_adaptive_verbosity` scored termination on `tokenizer.eos_token_id` (`<eos>`) but Gemma 4 closes a turn with `<end_of_turn>` (106). Corrected re-measure (`_resolve_stop_ids`): base_factual 48→**100**%, v7mc_factual 4→**100**% — **no collapse**; v7mc factual answers are correct + terse + terminate ("Paris"/"Au"/"1945"); the "degenerate loop" read earlier was junk *after* `<end_of_turn>`. The analogous D.1 cells were re-measured: no →4% collapse, only a modest open-ended-generation degradation (social 96→76%, creative 68→12%), adapter-dependent and off the V7-mc headline.	buggy: `EOS_v7mc_smoke.json`; fix: `mechanism_v8/EOS_verbosity_fixed.json`

Table 13: Deployment routes on the 198 base-wrong ARC items (the numeric view of Figure 2).

Deployment route	ARC accuracy (198 base-wrong)	Cost / item	Relation
Base letter-emission (1 forward, no CoT)	**0.970** (192/198; 2 unparsed as wrong → floor)	1 forward (a few decode tokens)	**matches SC-CoT; dominates the selector and adapter at equal cost**
Base self-consistency CoT (k=3)	**0.970** (192/198; k=5 identical → saturates at k=3)	~1179 generated tokens	ties letter-emission at ~1000× the cost
Read-only selector (Section 5.4)	**0.914**	1 forward (~1000× cheaper than CoT)	dominated by letter-emission; Pareto-dominates only the *adapter*
Contrastive adapter (cold-MC scoring)	**0.763**	1 forward	dominated by all three

A Extended mechanism detail and exploratory analyses

This appendix collects (Section A.1) the full mechanism-decomposition detail behind Section 5.8's verified negatives, moved out of the body for length, and (Sections A.2 and A.3) the exploratory single-seed / single-model analyses that extend the mechanism story but that the paper relies on for **no** claim. Sections A.2 and A.3 are exploratory (single seed / single model / not load-bearing).

A.1 Full decomposition for Section 5.8 (temperature negative, symbol channel, frequency control)

Not a temperature / null-space mechanism. A natural mechanistic hypothesis, given the readout framing, is that the offset acts like a confidence-regulation neuron: writing into the unembedding null space to rescale the residual-stream norm and thereby the logit temperature, scaling all logits without re-ordering the argmax [32]. A direct-logit decomposition refutes this for θ_{mc}. We split the offset's effect on the vocabulary into a uniform component (a temperature shift, parallel to the all-ones vector) and a ranking component (a re-ordering, perpendicular to it), and project the residual write onto the *genuine* temperature direction $d^* = \arg\min \|W_U r - \mathbf{1}\|$ (whose own logit image is 0.9988 uniform, confirming it is well identified). The offset's cosine with d^* is **0.023** — orthogonal to the temperature direction at chance level (the random-direction orthogonality baseline is $\sim 1/\sqrt{d_{\text{model}}} \approx 0.014$), and no single head is temperature-pure (per-head maximum uniform-shift fraction 0.58). (The raw logit-space uniform-shift fraction, 0.301, does *not* discriminate, because Gemma's unembedding is anisotropic and a random direction already produces ~0.24 ± 0.18 of apparent uniform shift; we therefore rest the negative on the d^* projection, not on the raw fraction.) The offset moves *differences* of logits, not their mean — it is a ranking edit, not a temperature edit — which closes a "the lift is just a confidence/temperature rescaling" alternative a reviewer could raise. **This is now confirmed causally.** A real-forward activation-patching test — the causal upgrade of the direct-logit reading — measures the offset at the scoring positions where the lift lives (the option-continuation tokens) and finds a **relative** temperature fraction of only **0.215** ($\text{temp}/\sqrt{\text{temp}^2 + \text{rank}^2}$, CI_{hi} 0.224 ≤ the pre-registered 0.35 kill-rule; in absolute fractions of the effect: temperature **0.177**, ranking **0.803**): the offset re-ranks, temperature is minority (the softmax-invariant uniform component, **0.4592** at the option positions, is excluded from the verdict). The causal cosine with the temperature direction, −0.149, is ~6× the direct-logit 0.023 and ~10× the random-orthogonality baseline (~0.014) — and of *opposite sign*: the linear attribution mis-stated the alignment in sign as well as magnitude, which is the expected signature of a first-order read of an off-axis causal component (whose first-order gradient is near zero, so its linear projection is dominated by noise terms that carry no sign information) rather than an anomaly of the measurement — yet the causal temperature fraction remains a clear minority (0.215 relative), so the negative rests on the causal measurement, not on the direct-logit cosine. The same logic is why we ran the frequency control's own causal cross-check rather than rest it on the attribution level (below): it comes back without a sign-flip, so the two Stolfo components dissociate under causal measurement. This under-reading of a causally effective, off-axis component by a *linear* (direct-logit) attribution is itself a documented pattern in the steering literature: for single-neuron control, first-order gradient attribution *anti-predicts* controllability precisely because effective controllers write **off the readout axis** and carry a near-zero first-order gradient, and a forward-only contrastive screen recovers the controllers that attribution misses [24] — so a linear read of our offset under-stating its causal alignment is the expected signature of an off-axis mechanism, not an anomaly. *Position matters:* measured instead at the next-token (generation) position the offset flattens the distribution more (temperature fraction 0.38, Δentropy +0.99, γ −0.59) — the generation-face flattening we connect to the format face below — so the temperature negative is anchored at the scoring positions, not the generation one.

The signpost: a content/symbol channel, not a lexical one. Direct-logit attribution over the 48 adapter heads finds **no clean lexical mechanism**: the tokens it *promotes* are diffuse and multilingual, with no coherent semantic theme. The one robust signature is on the *suppression* side — single uppercase letters and structural tokens (`B, S, P, T, C, L, Y, H, I, G, D, O, V, W` and punctuation), exactly the surface tokens of multiple-choice labels and turn formatting — the offset's logit-level correlate of the *format-preference* uncertainty that aligned-model calibration distinguishes from answer-decision uncertainty [17]. This is the readable face of the **format-preference channel** — the same generation-face flattening the causal test finds at the next-token position (above) — not of a termination collapse (that claim is retracted, Section 5.7) and not an explanation of the scoring lift, and it is an aggregate-of-heads property (a coherent sum across heads, not a per-head one). With temperature ruled out, the most promising unifying-mechanism candidate is the **content/symbol-binding channel** that MCQA mechanism work isolates: models first select a winning answer in *content* space and then bind or route it to the *symbol* to emit [39], a "correct-letter" routing studied at larger scale [23]. The decisive behavioral test — a TEXT-vs-LABEL scoring ablation (whether the lift survives permuting the option labels) — has now **run, and the result is consistent-with but not clean**. On ARC the label-arm (letter) lift exceeds the text-arm lift (Δ_{letter} +0.1575, CI [+0.087, +0.228]) and survives a headroom normalization (norm Δ_{letter} +0.17, CI [+0.088, +0.249]), consistent with a symbol-channel contribution — but the two permutation controls that would separate a *letter-specific* effect from a *generic symbol slot* are themselves ceiling-starved (caps_random base 0.78, numeric 0.97), so letter-specificity is **untestable** here. On TruthfulQA the letter arm is itself at ceiling (base 0.81), so its raw Δ_{letter} −0.29 is a **headroom artifact** — the task is *uninformative* rather than contradictory (a point we surface only after normalizing; the raw sign looked opposite). The symbol channel is therefore **consistent-with, not a clean mechanism**: the lift already lives in the text-face arm (which reproduces +0.3275 ARC / +0.3575 TQA on its own), so permuting the labels does not collapse it. We report this as a signpost, not a named mechanism. This suppression signature is now confirmed *causally on real generation positions* — an nnsight patch of the offset over 50 gsm8k traces (26 non-terminating loops + 24 clean-wrong), measuring the base→adapter logit delta per generated token: of the tokens the offset most suppresses, ∼**91**% **are structural** (punct/symbol), a ×**3 enrichment** over the vocabulary base-rate — the same structure-suppression face found at the scoring positions, now on the generation face — and the structure share **grows mildly along the generation** (slope +0.04, slightly steeper in the loops). But the signature is strong only in the tail (diffuse in raw mass) and its growth is sub-threshold, so it does **not** license a "commitment-machine" reading of the reasoning loss; we leave that frame as an undecided conjecture, not a mechanism. A control that the promotion is not a token-frequency artifact — the *other* Stolfo component (token-frequency neurons) — has now been **re-derived, persisted, and closed** (`derive_freq_control`, reusing the same aggregate-write construction, ‖agg‖ 100.28). Over the full wikitext-103 training corpus (121.6M tokens, 112.8k of the 262k vocabulary observed) the recovered frequency direction d_{freq} is well identified — its logit image reproduces the target log-frequency at $r = \mathbf{0.906}$, past the pre-registered 0.9 self-gate — and the offset's cosine with it is **0.0128**, below the 3×-null gate (0.041) and at the null floor ($1/\sqrt{d} \approx 0.0136$). The identification rose monotonically with corpus size ($r = 0.855 \rightarrow 0.887 \rightarrow 0.897 \rightarrow \mathbf{0.906}$ at 5M/20M/40M/121.6M tokens), crossing the gate only at the full corpus. So the offset is **not** the token-frequency component of a Stolfo mechanism either — a closed control, not merely consistent-with. **The causal-patching cross-check has now run** (the same protocol as the temperature one: real forward, ARC, n=400, scoring positions; the six previously published decomposition scalars reproduce to $\Delta = 0.00$, so the frequency arm is strictly additive). Projecting the real base→adapter logit delta onto the mean-centered frequency signature gives cosine **+0.0059**, CI [−0.0027, +0.0142] — **same sign** as the direct-logit 0.0128, ×0.46 its size, and crossing zero — against the temperature's flip to −0.149 (CI [−0.158, −0.140], ×6.5). The per-item view is sharper than the means: the frequency projection's mean *absolute* size is comparable to the temperature's (0.120 vs 0.177), but its **sign is inconsistent across items (66% positive, vs 99% negative for temperature, with γ<0 in 99.3%)** — the two components dissociate not in how much they load but in whether they load *coherently*, and only a coherent component can drive a lift that is systematic across items. **One declared limit:** the frequency direction orthogonalized against the base logits (the part of log-frequency not already shared with the temperature channel — a quantity with no direct-logit counterpart, so it neither confirms nor contradicts the 0.0128) does carry a small but non-zero causal load, cosine **+0.0298**, CI [+0.0221, +0.0374]: ~15× the vocabulary-space null ($1/\sqrt{V} \approx 0.002$) but ~1/5 of the temperature component and ~1/27 of the ranking one. We therefore state the negative at its true strength: token frequency is ruled out as *the mechanism*, not as a trace contaminant of the offset's logit image.

A.2 Off-workspace, not workspace-broadcast (J-lens + recall-tail) — exploratory, single seed

Off-workspace, not workspace-broadcast: the off-axis push is an automatic-channel edit, not a global-workspace one. If the off-axis push is a shortcut that the cold-scoring readout reads but generation does not (Section 5.1), a natural question is whether it enters the model's *global workspace* — the broadcast subspace whose contents a Jacobian-lens ablation degrades — or acts off-workspace as an automatic channel. The workspace account [16] predicts the second: it reports that multiple-choice scoring is *not* degraded by J-space ablation (automatic) while GSM8K-CoT is robust because the model externalizes what it would otherwise hold in the workspace — both of which map one-to-one onto θ_{mc}-lifts-MC-but-not-CoT. We fit a Gemma Jacobian-lens (`anthropics/jacobian-lens`, 114 prompts, above its ~100-prompt saturation, source layers → L58) and test whether θ_{mc} behaves like a workspace-broadcast direction. Four independent reads converge on **off-workspace**. (i) An MLP-gain kill-switch — testing whether the MLPs amplify θ_{mc} the way they amplify broadcast directions — returns gain ≈ 1.0–1.4×, indistinguishable from a random vector (~1.0×) and far below a J-lens broadcast reference (`jlens_dir_highref` 2.3–4.0× at L37–40); θ_{mc} is not amplified like a workspace direction (the kill-switch did not fire). (ii) The J-space fraction of θ_{mc} equals a norm-matched random vector's (excess ≈ 0). (iii) At option positions the offset inserts *no* answer-letters (insertion rate 0.0 at every layer — the +35 is option-text, not letter, consistent with Section 5.9), yet it *does* rewrite the late readout (recall@25 falls from 0.98 at L30 to 0.34 at L49, strongest in the offset's own layers). (iv) The tokens it rewrites the readout toward are **structure, not knowledge** — bare capitals, punctuation and markup, multilingual fragments, zero semantic/answer tokens — the same structure-suppression signature the causal patch found. This read is resolved *per-position* on the real base-vs-adapter activations (the recall-tail): across the tail layers where recall@25 falls (L38→L55), the adapter-only inserted ids are **60–88% structure/format** (bare capitals, quotes/punctuation, non-latin script) with **zero answer-tokens at every one of the twelve layers**, and the non-structural remainder is generic multilingual function-words (`and`, `de`, `own`, `ability`, `same`), *not* answer-knowledge — so even the 60% floor under-states the no-answer-knowledge share. So the offset perturbs the late readout toward output *format*, not via workspace broadcast and not toward the answer: an automatic-channel edit, which is the mechanistic shape a direction that lifts cold scoring but does not transfer to reasoning should have. We separate what is measured from what is inferred here: the load-bearing E2 signals are occupancy and recall@25 (the readout *is* rewritten late, recall@25 0.98→0.34); the "zero answer-*letters* inserted" figure is a designed null-control (`run_p10_pod.sh` expects ~0 for A/B/C/D insertion, consistent with Section 5.9's option-text-not-letter finding), not a discrimination of off-workspace from an answer-directed push. The `structure_fraction` (60–88%) is biased toward structure by the top-50 cut (item-specific content tokens have low counts and fall outside). A gold-option-text occupancy metric (top-25 vs the gold continuation tokens) would firm up this read; it is flagged as the Section 5.8 follow-up, not run.

The J-lens sanity gate is marginal, and we scope the claim to what it licenses. The fitted lens clears its mandatory sanity gate — recovering the model's own next-token from intermediate residuals — only *marginally*: at just 1 of 18 fitted layers (L55 @ 0.50, the layer nearest the L58 target), 0.0 elsewhere. We diagnose this as the known Jacobian-lens limit, not a broken fit: recovery is governed perfectly by distance-to-target (Spearman(recovery, distance) −1.000; every scale/shape metric of the averaged Jacobian grows monotonically toward the target — $||J||_F$ ρ +0.996, mean-diagonal ρ +1.000), and far from the target the averaged Jacobian collapses toward rank-1 with near-zero trace (one linear map cannot bridge many nonlinear layers). The load-bearing read (iv, the structure-not-knowledge readout) **reproduces at L55, the one sanity-valid layer** (90% structure, zero knowledge/answer tokens — the same class as the marginal layers L30–49). We therefore scope the workspace claim precisely: *off-workspace is the convergent read across the four probes above and an independent causal channel (the nnsight patch, Section 5.8), and the J-lens is quantitatively valid near its target, where the direction read reproduces* — **not** that the J-lens is globally valid. A per-source-target re-fit would harden the early-layer reads; more prompts would not (the marginality is structural, not sampling noise). Exploratory, single seed.

A.3 Dose-window efficiency candidate — exploratory, single seed; steering ≠ baked

A dose-window efficiency candidate — a positive worth chasing, but gated. The single-magnitude activation-scale axis that dissociates the collateral cost (Section 5.7) surfaces an unexpected *positive*: at intermediate doses the offset **shortens** generative CoT without breaking it. Conditioning on the intersection where both the base and the scaled offset answer correctly (n=200 gsm8k), scale 0.75 cuts generation length **−34.7**% while the real arithmetic stays intact — verifiable calculation steps fall only **−7.2**% against a −19% drop in «. . . » annotation markers and a −52% drop in prose, with 98% of remaining steps still valid, and a qualitative arbiter confirms complete reasoning even where the «» markers are gone. The genuine shortcut/breakdown — calc −18%, step-omission, creative-degeneration loops — appears only at full strength s=1.0, where accuracy collapses to 0.815. So *inside* gsm8k there is a threshold dissociation: an intermediate dose prunes prose while preserving the computation — the same phenomenon as the dose-separable collateral cost (Section 5.7), read on a different metric. We flag this as a **candidate, not a claim**: it is measured on the *injected, scaled* steering offset (steering ≠ baked), on a single seed, and whether a trainable adapter inherits the efficient regime is exactly the open question that fork (b) above leaves standing.